

MAGNETOM

Troubleshooting Guide System

RF System

Avanto
Espree

Document Version

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1.1 General

In this section, you will find the **strategy** and **test procedures** for troubleshooting the RF system.

One of the basic prerequisites for excellent image quality in an MR system is an accurate and stable RF system.

However, due to the complexity of an MR-system, image quality is influenced by a great variety of components and functions, such as

- Image generating components (e.g. static magnetic fields, RF system, gradient system)
- Control systems (e.g. Host, AMC, MARS)
- Communication systems
(e.g. CAN bus, general data transfer bus systems)
- Image acquisition, processing and archiving system
(e.g. Imager, MARS, archiving media)
- Software performance

The main tools for RF troubleshooting are the **RF tests** available in the **"TestTools"** service software platform.

The RF test includes communication tests, power tests (RFPA) and measurements of stability, linearity and signal levels of RF components using loop tests, which incrementally include more components on the TX/RX path. If one test fails, a diagnosis shows you the FRUs responsible.

Additional hardware tests are described in the **"Procedures"** section.

The **"Replacement of parts"** manual describes the necessary procedures, adjustments and final checks when replacing an FRU.

The following tables are designed to provide the FSE with shortcuts, describing the symptom, the appropriate test, and the respective FRU in the RF system.

1.2 Proposed sequence of test procedures - FRU-based

The following tests are recommended to verify the function of a specific Field Replaceable Unit (e.g. in case of problems after FRU replacement).

Tab. 1 RFSU

RFSU	
FRU	Test Procedures
Synthesizer D10 (73 88 734) D120 (7T/3T) (10018540) Avanto: >26124 Espree: > 30292	the synthesizer is the reference for the RF system. It can be only tested with MR-measurements, e.g. QA/ Stability Check (see Replacement of Parts description (→ Synthesizer / M6-020.841.14)).

RFSU	
FRU	Test Procedures
Transmitter D111 (100 92 367) (replaces D101 and D10/120)	Test Tools / MOD / REC Test or TestTools/ TX/ RX Tests

RFSU	
FRU	Test Procedures
Modulator D101 (73 89 062) with MKO: Modulator 7T (100 18 301) (Usable for 1.5 / 3 T or other sys- tems)	Test Tools / MOD / REC Test or TestTools/ TX/ RX Tests

RFSU	
FRU	Test Procedures
Receiver D112 (73 89 047)	Test Tools / MOD / REC Test or TestTools/ TX/ RX Tests
Receiver D122 (10276813)	
Avanto: >26410 Espre: >30439	
Power Supply (30 95 783)	MR image

RFSU	
FRU	Test Procedures
Backplane D20 (73 89 138)	MR image
RFSU frame (73 91 001)	MR image
ACC internal cable	Test Tools / RFPA Test
RFSU internal cable (e.g., 73 91 456)	TestTools/ MOD/REC Test or TestTools/ TX/ RX Tests

Tab. 2 RFAS

RFAS	
FRU	Test Procedures
TALES (73 91 886)	see Replacement of Parts description (→ Final checks / M6-020.841.14).
BCCS_63 (Avanto) (47 53 229)	
BCCS_63P (Espre) (100 17 953)	

RFAS	
FRU	Test Procedures
RCCS D11 (73 89 013) Avanto: <27081 Espree: <30785 RCCS_2 D11 (102 77 196) Avanto: >27080 Espree: >30784	Test Tools / RCCS tests maximum: 3.5 hrs (with SW VB17A/ VD11A or higher: 20 min)
TAS_3TM (73 92 058) Avanto: <27127 Espree: all TAS_CM (102 77 008) Avanto: >27126 Espree: not used	- Test tools/ RFPA test or TestTools/ Fast/ TX - MR measurement with local TX/RX coil (e.g., Extremity coil)
SAMI (73 83 123) (only for systems with MKO)	MR measurement with Body Coil and local TX/RX coil.
RF cable between TALES and Body Coil	Test Tools / BC Test or TestTools/ Interactive/ Cable BC

Tab. 3 RFIS

RFIS	
FRU	Test Procedures
RFIS Mother- board (73 89 195)	- TestTools/ RFIS Test or TestTools/ Interactive/ LC Plug x - Test Tools / BC Test or TestTools/ Normal/ TX
DYSCON_BC (73 89 377)	Test Tools / BC Test or TestTools/ Normal/ TX
DYSCON_16 (73 88 775)	TestTools/ RFIS Test or TestTools/ Interactive/ LC Plug x

RFIS	
FRU	Test Procedures
CAN_AND_CODE_CTRL (73 89 351)	TestTools/ RFIS Test or TestTools/ Interactive/ LC Plug x
Power supply E4A (30 95 700)	- Check Error LEDs on RFIS - TestTools/ RFIS Test or TestTools/ Interactive/ LC Plug x - Test Tools / BC Test or TestTools/ Normal/ TX

Tab. 4 RFPA

RFPA	
FRU	Test Procedures
RFPA (Avanto/ DORA) (47 53 062)	Test Tools/ RFPA test or TestTools/ Fast/ TX/ Max Power
RFPA (Espree/ PORA) (73 92 470)	
Systems with MKO: (Avanto/CORA) (73 83 339)	see Replacement of Parts description (→ Final checks / M6-020.841.14).

Tab. 5 Body Coil

Body Coil	
FRU	Test Procedures
Body Coil Avanto (81 20 540)	Test Tools / BC Test or TestTools/ Normal/ TX/ BC Tests
Body Coil Slim 70 Espree (10048071)	

Tab. 6 Local Coil

Local Coils	
FRU	Test Procedures
Local Coils	▪ Check Error LEDs on RFIS and RCCS ▪ TestTools/ RFIS Test or TestTools/ Interactive/ LC Plug x

1.3 Proposed sequence of test procedures for RF problems

Proposed sequence of test procedures for RF problems RCCSTest (interactive) may be run overnight.

No	Test	Duration	Reference
1.	Run TestTools Communication Tests or TestTools/ Fast Status and Control	3 min.	TestTools AMC/RF
2.	Run TestTools AMC tests or TestTools/ Fast Status and Control	30 min. or 10 min	Online help TestTools/RF
3.	In case TX path/ TX power problems: Run RF tests (functional) or TX/RX tests	60 min.	Online help
4.	In case of local coil problems/ receive-path problems: Run RF tests (RCCS/RFIS) or TestTools Interactive RF Tests	180 min. or 40 min	
5.	Perform TuneUp procedures (Tuning Calibration; BC Tuning; RF Characteristic; Coil Power Losses or BC Power Losses; Receive-Path Calibration/ RX Gain Calibration)	dependent on procedure	TuneUp Online Help



The RCCSTest (interactive) may be run overnight.

1.4 RF-related hardware errors, system control errors

RF-related hardware errors, system control errors			
No	Symptom	Service Action	Reference
1.	Specific Error Messages Specific defective components Adjustments/measurements fail	Check and search corresponding Error messages in the EventLog window (from the Service Software UI). Clicking the ID of the error message opens an additional window with further information, such as explanations, service actions and additional information (if available). Check error LED's, power supplies, fuses. Check communication cables (FOCs, optical transmitter and receiver), use TestTools AMC tests or Status/ Control tests. Interchange like components with one other (e.g. PCI_TX and PCI_MON) Perform specific RF tests	Functional description of corresponding components Diagrams AMC tests or Status and Control tests Online help RF procedures
2.	Communication errors CAN errors Software hang-up	Run TestTools AMC: PCI-CAN, LWL_INT RF: Comm/Status or TestTools/ Fast/ Staus and Control (CAN units of the RF system are: RCCS, RFIS, RFPA) Reset RFSU by switching off/on fuse ACC/F23 or by disconnecting/reconnecting FOC/U4 from/to PCI-TX.	TSG Control TestTools Online-help
3.	No/low RX signal, noise image only	Check cables/connection in the RX path Check signal level of RF components with scope	RF procedures TestTools/RF/RCCS/ RFIS or Interactive tests
4.	No/low TX signal	Run TestTools/RF Check signal levels of modulator, RFPA(PF) with scope Run TestTools/RF/Cable (Interactive)	RF procedures TestTools/RF/Cable

1.5 RF-related image quality problems

RF-related image quality problems (e.g specific QA procedures, such as Coil Check, CalcArtifacts, Stability Check fail and/or in case of general image quality complaints.

No	Symptom	Service Action	Reference
1.	Ghosting in phase decoding direction Independent of orientation	Run TestTools/RF Tests: MOD/REC; do a check of the stability	TestTools RF Online help
2.	Low S/N	Run TestTools RF tests: MOD/REC, RCCS, RFIS; check the results of stability and linearity evaluations. Eliminate possible spike sources.	TestTools RF Online help
3.	Slice profile broadening	Run TestTools RF tests; check the result of linearity evaluation. Check modulator SSB 1 with scope and RHP. (→ RFSU / Page 17)	TestTools RF RF Procedures
4.	IQ problem with local coils depending on coil plug used	Run TestTools RCCS/RFIS, QA/Coil Check on various coil plugs (→ RCCS Procedures / Page 28), (→ RFIS Procedures / Page 34)	TestTools RF RF Procedures
5.	IQ problem with local coils independent of coil plug used	Check local coil/ use another coil Run TestTools RCCS/RFIS; MOD/REC; (→ RCCS Procedures / Page 28), (→ RFIS Procedures / Page 34)	TestTools RF Online help

1.6 Error messages

The error messages are listed in the (→ SKB, <https://skb.healthcare.siemens.com/>).

2.1 General

This section describes procedures for troubleshooting components (FRUs) of the RF system, such as voltage level and pulse form measurements at the **RFSU** (modulator, synthesizer), **RFPA**, **RFAS** (TALES, RCCS, RFIS) and local coil plugs at PTAB. Additional procedures use the **SeSo/Tune-up functions** and **Seso/Testtools**. For **special problems (especially RFPA/ RF power)**, troubleshooting methods are described.

2.1.1 Required Tools

- Function Description RF system (FUN)
- Diagrams RF system (Avanto and Avanto Dot)
- Set of standard tools
- 50 Ohm/2W BNC feedthrough termination resistor (7760457)
- QLA T connector (30 58 971)
- Service cables QLA-BNC (87 33 446)
- Connector BNC-Banana plug (3025749)
- Digital multimeter, type Fluke 187 or equivalent
- Oscilloscope, 200MHz bandwidth
(e.g. Fluke Scopemeter 199CM)
- Service Plug kit, Avanto/ Espree (7715720)



To stabilize the temperature in the control cabinet, please ensure that the system is powered on for at least 20 minutes before taking any measurements and/or running any functional or interactive test from the Service Software procedures (TestTools/RF tests).

2.2 RFSU

Measurement of:

- Modulator SSB1 signal
- Synthesizer 10 MHz signal

2.2.1 SSB1 - Signal

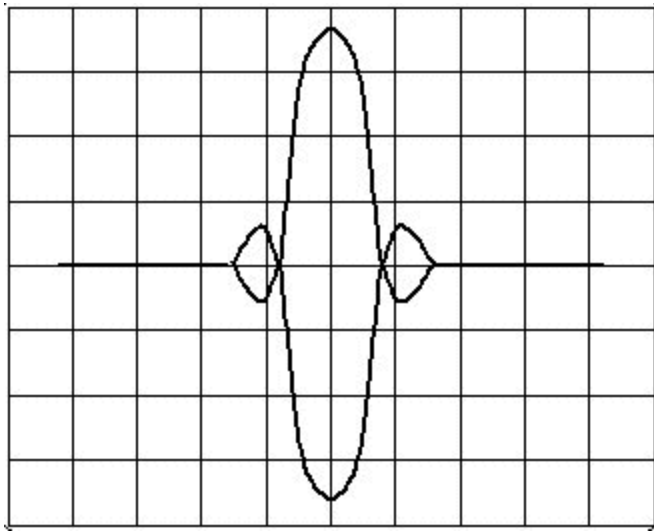
2.2.1.1 Prerequisites

- MR system ready for measurement
- Body phantom with loader positioned in the magnet center.

2.2.1.2 Performing the Procedure

- Start the Service Software platform. Start TestTools. Click on "Tools" in the header bar. In the Tools platform, register the service patient. Exit the platform with "Save".
- Reduce the service platform to an icon.
- From the examination manager, Siemens/Sequence, Region/ServiceSequences/DefaultProtocols, fetch the sequence **rf_pulse**.
- Open the sequence. Adjust the following parameters: Click on **System** and **Transmitter/Receiver**: Set RF amplitude sSRF01[1H] to 100 V. Click **Contrast**: Set measurements to 100. Click **Sequence** and **Special**: Set RF Pulse Length to 2 ms. Set Loop Mode to TX only. Set Pulse Type to Sinc.
- Connect the SSB1 signal on the modulator via QLA-T-connector to the RFPA and to the scope CH1 via a feedthrough 50 Ohm terminator. Connect AMC/PCI-TX/ X8 output OS-ZI_TRIG (0) or AMC_IO/ X8 as an external trigger to the scope.
- Start the measurement by clicking on Apply.
- Oscilloscope setting: CH1: 0.01 V/div, Time: 0.5 ms/div, ext.Trigger (use X8 at AMC/ PCI-TX or AMC_IO/ X8 as external trigger source).

Fig. 1: Sinc pulse



- Monitor the SSB1 signal (constant sinc-pulse) on the oscilloscope.
 - » A symmetrically-shaped sinc signal must be continuously present without voltage drop-off or overshoots.
- Abort/stop the measurement
- Restore the original system configuration.

2.2.2 10 MHz - Signal

2.2.2.1 Prerequisites

- MR system is on.

2.2.2.2 Performing the Procedure

- Remove the QLA connector from AMC/PCI-TX/X7 or AMC_IO/ X7 and connect this cable via the QLA T-connector to the oscilloscope via a feedthrough 50 Ohm terminator.
- Oscilloscope setting: CH1: 0.1V/div, Time: 50 ns/div, Trigger CH1.
- Monitor the 10 MHz signal on the oscilloscope.
 - » A sine-shaped signal with an amplitude 0.42V or more (>2.5dBm) must be displayed.
- Restore the original system configuration.

2.3 RFPA

2.3.1 Directional Coupler Signal PF

2.3.1.1 Performing the Procedure

- Remove the QLA connector from the input PF of receiver #1. Connect a QLA T connector to input PF input of receiver #1. Connect the cable from PF output of RFPA (forward power directional coupler of RFPA) at one end of the T and the scope CH1 via feedthrough 50-Ohm terminator to the other end. Connect the AMC/PCI-TX/X8 output OSZI_TRIG (0) or AMC_IO/ X8 as an external trigger to the scope.
- Use the same service sequence as for the SSB1 signal test.
- Oscilloscope configuration: CH1: 0.01 V/div, Time: 0.5 ms/div, ext.Trigger.
- Monitor the signal PF (constant sinc-pulse) on the oscilloscope.
 - » A symmetrically-shaped sinc signal must be continuously present without voltage drop-off or overshoots.

The attenuation of the directional coupler is approx.70 dB at 63 MHz.

- Abort the measurement.
- Restore the original system configuration.

2.3.2 Tripped-off Circuit Breaker F11 (DORA/ PORA)



CAUTION

The RFPA can be seriously damaged by improper power switching directly at circuit breaker F11/ACC !

Switching-on the RFPA supply voltage directly at circuit breaker F11/ACC can cause irreversible damage to the RFPA soft-start unit!

- » Before operating circuit breaker F11/ACC, always first switch-off the MR system to stand-by.
- » Then, wait at least 3 minutes before switching circuit breaker F11/ACC.
- » Never switch power to the RFPA directly at circuit breaker F11/ACC !

Correct RFPA power switch-on sequence (after circuit breaker F11/ACC has tripped-off):

- Switch the MR system to **Standby** at the Alarm Box.
- Wait at least 3 minutes to discharge RFPA voltages..
- Switch-on RFPA circuit breaker F11/ ACC..
- Switch the MR system **ON** at the Alarm Box.
- **Investigate the cause**, measure the RFPA supply voltages as follows:

Tab. 7 AC supply voltage at DORA/ PORA

Point of measurement	L1- L2	L2- L3	L3- L1	Maximum charged deviation (max-min)
Spec	166,3 V +/- 8 V			0,8 V
Result				

Point of measurement	L1-N	L2-N	L3-N	Maximum charged deviation (max- min)
Spec	96 V +/- 4 V			0,5 V
Result				

Tab. 8 Testing range

Point of measurement	Spec	Measurement value	Testing range	
N- PE	117 V +/- 2 V		DC	
	17 V +/- 2 V		AC	



Voltages are measured at the specified AC line voltage of the MR system (see Planning Guide).

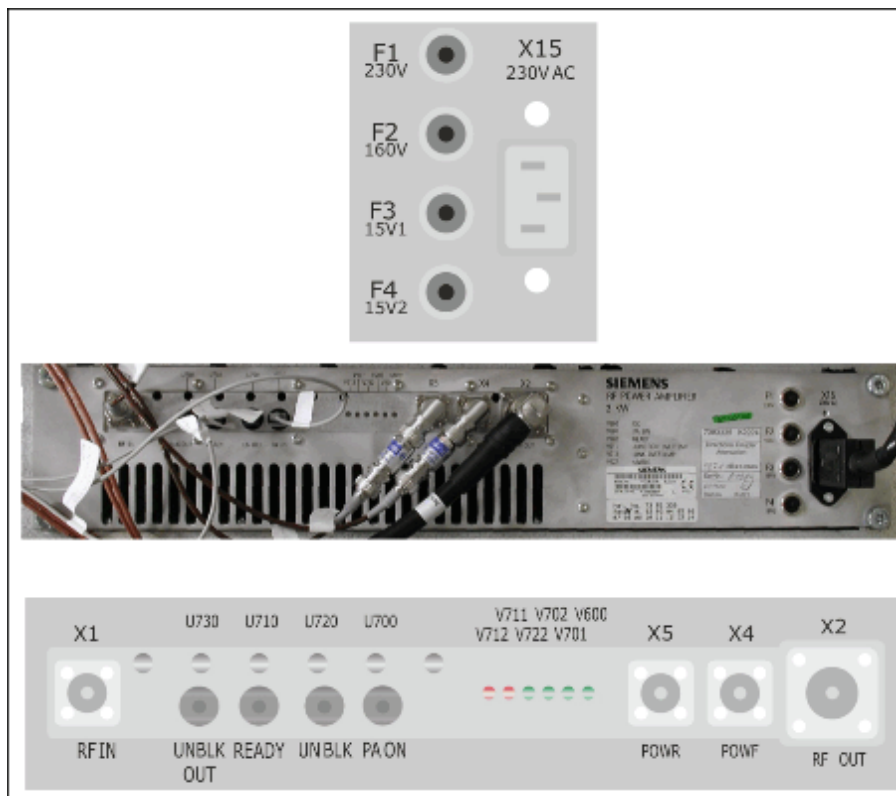
Line voltage fluctuations can cause deviations from the expected results.

2.3.3 CORA Shut-downs



The CORA RFPA (material number 7383339) is part of the Multinuclear Spectroscopy option.

Fig. 2: CORA RFPA front view



In case the status of the CORA RFPA is not ok (e.g. amplifier shutdowns for unknown reason), perform the following checks:

- Check if the three green LEDs (V600, V701, V702) at the CORA are **ON** and green LED V722 is flashing when a sequence is running:

Tab. 9 LEDs at CORA

LED	Description
V600 "DC" (green)	ON indicates internal power supply unit OK
V701 "PA ON" (green)	ON indicates amplifier is being enabled (no RFSWD errors). The PA_ON signal comes from IOP/ LWL_INT board via FOC connector U700 at CORA.
V702 "READY" (green)	ON indicates amplifier start-up completed and unit now ready.
V722 "UNBLK" (green)	ON (flashing) indicates unit being unblanked (i.e., when transmitting).

LED	Description
V711 "JUNCTION OVERTEMP" (red)	ON error indicates transistor junction temperature too high (non-latched).
V712 "SINK OVERTEMP" (red)	ON error indicates heat sink temperature too high (non-latched).

- In case a **CORA shuts-down already at sequence start**, check:
 - if the max. pulse amplitude is too high
 - if the RF-pulse duration is too long (wrong sequence parameters, customer-programmed sequence?)
 - for correct timing of UNBLANK signal. Swap FOC "UNBLK" CORA with DORA. If the error at CORA disappears, replace "UNBLANK" FOC cable from IOP/ LWL_INT board.
 - for interrupted PA_ON signal at FOC U700 (broken FOC?, IOP/ LWL_INT defect?)
 - remove the CORA RFPA input (X1, RF_IN) and start sequence again. If the error persists, the shut-down is NOT caused by too high max. pulse amplitude but may result from a too long pulse duration or interrupted PA_ON signal at FOC U700.



CORA shut-downs at sequence start are usually NOT caused by RFPA over temperature and are NOT caused by too high average power.

- In case a **CORA shuts-down after some measurement time**, check:
 - the RFPA cooling water temperature and flow under **SeSo > Magnet&Cooling > Cooling Status**.
 - if the average power demanded by the sequence is too high (wrong sequence parameters, customer-programmed sequence?)

2.3.4 Error LED's RFPA PORA (Espre)

2.3.4.1 Configuration of LEDs in front

	V 27	V 26	V 25	V 24	V 23	V 22	V 21	V 20	V 19	V 29	V 18	V 17	V 16	V 15
V 14	V 13	V 12	V 11	V 10	V 9	V 8	V 7	V 6	V 5	V 28	V 4	V 3	V 2	V 1

2.3.4.2 Description of LEDs

LED/ color	Signal Name	Description
V1 green	L1	AC voltage L1
V2 green	L2	AC voltage L2
V3 green	L3	AC voltage L3

LED/ color	Signal Name	Description
V4 yellow	CAN_TX	CAN TX active, PORA in send mode
V5 green	READY	Capacitors charged, RFPA supply voltage ready
V6 yellow	STARTUP	Power on signal to RFPA, capacitors charging starts
V7 yellow	UNBLANK	Unblanking signal, RFPA is biased
V8 red	DISSIPATION 1	Dissipation error in module RFPA 1
V9 red	RF_OVER 1	RFPA 1 output power too high
V10 red	TJ_FAULT 1	RFPA 1 junction temperature too high
V11 red	N_OVERCURRENT	Peak or average RFPA 1 + RFPA 2 current error reached
V12 red	C_V_OVER	Voltage at capacitors too high
V13 red	U_R_OVER	RFPA supply voltage too high
V14 red	U_R_FAULT	RFPA supply voltage too low
V15 red	5V_FAULT	5V supply error
V16 red	15V_FAULT	15V supply error
V17 red	N_15V_FAULT	N15V supply error
V18 yellow	CAN_RX	CAN RX active, PORA in receive mode
V19 yellow	N_RFPA_ENABLE	RFPA disabled by U6
V20 yellow	CAN_OFF	CAN_OFF by SLIO
V21 red	DUTY_CYCLE	Duty cycle error
V22 red	DISSIPATION 2	Dissipation error in module RFPA 2
V23 red	RF_OVER 2	RFPA 2 output power too high
V24 red	TJ_FAULT 2	RFPA 2 junction temperature too high
V25 red	OVER_TEMP	Water cooler surface temperature is too high (> 35 degree centigrade)
V10, V24, V25	Combined with CAN error message over_temp	- Cabinet cooling problem - Overdrive of RFPA - Surface temperature of water cooler < 35 degrees centigrade
V26 red	50_OHM DISS	Dissipation power in internal 50 Ohm termination too high
V27 red	50_OHM_VSWR	Input matching at internal 50 Ohm termination too high

LED/ color	Signal Name	Description
V28 yellow	PA_ACTIVE 1	RFPA 1 peak power > 1kW
V29 yellow	PA_ACTIVE 2	RFPA 2 peak power > 1 kW

2.3.4.3 Description of Error Messages

Error Message	Category	Possible cause
Temperature	Overload	The temperature of the heat sink is too high. The reasons could be a lack of cooling water or the junction temperature is too high after overloading the power stage transistors. It could also be a cabinet cooling problem.
Duty cycle	Overload	RF-pulse longer than 100 ms or pulse interval shorter than 0.1 ms.
Current	Overload	Mean or peak current from power supply unit too high.
RF_Power	Overload	$U_{\text{forward}} + U_{\text{reflected}} > U_{\text{nominal}}$ (under 50 Ohm condition)
Voltage	Overload or Fault	Missing supply voltages caused by failure of mains voltages or defective power supply unit.
Fault	Fault	Sum error message, caused by defect (defective component) in the RFPA.



After an error of the "Overload" category, the RFPA can be restarted again after eliminating the cause. The RFPA will work again within specifications.

2.4 RFAS

2.4.1 TALES 500 mV Reference Voltage

2.4.1.1 Prerequisites

MR system is on and in idle state.

2.4.1.2 Performing the Procedure

- Connect the cable from X7/TALES and a DVM via QLA T-connector to the CV input of the modulator.
 - Disconnect FOCs U1, U2 and U3 from AMC/PCI-TX and connect the FOCs to the outputs U1, U2 and U4 of AMC/PCI-MON.
 - Switch off F23/AMC/RFSU; switch it on again after a few seconds.
 - » During automatic boot of the AMC, the reference voltage of TALES can be measured with the DVM.
- Nominal voltage: 0.475 to 0.525 VDC.

2.4.1.3 Checking the FOCs at TALES

(If the reference voltage cannot be measured in the preceding test)

- Remove the MH_SEL (U1 - U3) from TALES.
- Disconnect FOCs U1, U2 and U3 from AMC/PCI-TX and connect the FOCs to the outputs U1, U2 and U4 of AMC/PCI-MON.
- Switch off/on F23.
 - » During the boot/self-test of AMC, all three FOCs light up.

The following table shows the selected TALES output channel depending on the select signals MH_SEL.(L= light OFF, H=light ON)

Tab. 10 TALES MUX Control - CV signal / Offset

SEL2	SEL1	SEL0	TALES	Information from X7/X8 to MOD/CV and MOD/OFFSET
L	L	L	X1 - X3	U_{forward} (BC 0-degree-system)
L	L	H	X1 - X3	$U_{\text{reflected}}$ (BC 0-degree-system)
L	H	L	X2 - X4	U_{forward} (BC 90-degree-system)
L	H	H	X2 - X4	$U_{\text{reflected}}$ (BC 90-degree-system)
H	L	L	X5 - X6	U_{forward} (Local TX coils)
H	L	H	X5 - X6	$U_{\text{reflected}}$ (Local TX coils)
H	H	L	X8	Offset voltage (used for correction)
H	H	H	X7	Reference voltage

- Restore the original system configuration.

2.4.2 TALES CV - Signal

2.4.2.1 Measurement of the CV Signal with a scope

- Check the correct function of TALES.
- Identify additional RF interferences and errors in the TALES modulator cable (CV input).
- Check the correct connection of FOC control lines MH_SEL U1, 2, 3 at TALES.

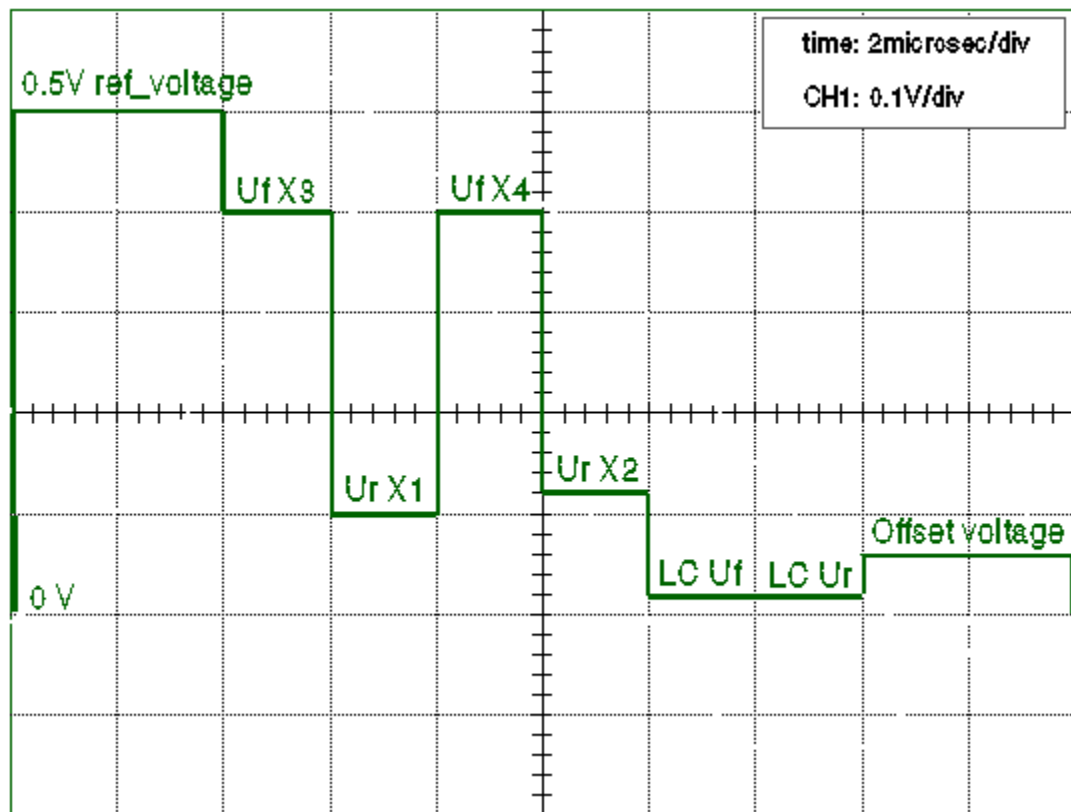
2.4.2.2 Measurement Configuration

- Using a QLA T-connector, connect the CV signal from TALES and the scope with the RFSU/modulator.
- Set scope CH1 to 0.1 mv/div and time scale to 2 μ s/div.
- Position the body phantom and loader in the magnet; register a suitable test patient.
- Load and open the service sequence rf_pulse: Select/check the sequence parameter cards:
- Resolution: Base resolution: 128
- Contrast: TR = 101ms, TE = 1ms, Measurements 20 (4:19 min measurement time)
- System, Transmitter/Receiver, Transmitter amplitude 200V (the pulse amplitude must be entered for each new measurement)
- Start the measurement. The following example of a waveform corresponds to a normal working TALES and intact signal line from TALES to the modulator CV input.
- Perform the measurement with different transmitter amplitudes and check the corresponding waveform changes on the scope. Alternatively, you can use a normal imaging sequence, e.g. se_15b130.

2.4.2.3 Waveform Display on scope

Under the measurement conditions described, the chronological sequence of the waveform segments is always the same. However, the display on the scope, i.e. starting point, depends on the trigger conditions. As a point of orientation, use the 0.5V reference voltage segment.

Fig. 3: Sequence of waveform segments



The signal heights of forward and reflected voltages are proportional to the transmitter voltage used.

MR system Avanto (TX to BC; transmitting to BC)

Tab. 11 CV signal / Offset

Signal from TALES	Duration (μ s)
Offset voltage	4
0.5V reference voltage	4
Forward voltage X1-X3 (BC 0-degree system)	2
Reflected voltage X3-X1 (BC 0-degree system)	2
Forward voltage X2-X4 (BC 90-degree system)	2
Reflected voltage X4-X2 (BC 90-degree system)	2
Forward voltage X5-X6 (LC TX)	2
Reflected voltage X6-X5 (LC TX)	2

2.4.3 RCCS Procedures

- Check the LEDs on the RCCS, if RCCS-related error messages occur.

2.4.3.1 Description of LEDs on RCCS

Tab. 12 LEDs and FOC connections on RCCS

LED/ FOC	ON	OFF
P5V red	+5V missing	+5V OK
P15V on Green	+15V OK	+15V missing
P15V red	+15V missing	+15V OK
N15V red	-15V missing	-15V OK
FPGA Yellow	Firmware FPGA loaded	Firmware FPGA not loaded
CCLK Yellow	Internal clock enabled	SLIO in sleep mode (internal clock disabled; e.g. during sequence on)
CTOG Yellow	Communication OK, SLIO awake	Communication NOK (may be in sleep mode; e.g. during sequence on)
SFAULT Yellow	All supply voltages are OK	At least one of the supply voltages is missing
V1 V64 red	The corresponding local coil preamplifier 10V supply is missing or short in cable or coil plug	All 10V supply voltages are OK
U1/U3 yellow CAN1, CAN2	CAN Bus active	CAN Bus not active
U5 GAIN Yellow	low gain	high gain
U6 - U9 (SEL_0 -SEL_3) Yellow	FOC lights up = 1 (Binary coded setting number(	FOC dark = 0

2.4.3.2 RCCS Tests



The Service Plug is used for the RCCS plug1 - 10 tests and the 10 VDC preamplifier test.

General

The TestTools/RF/RCCS tests(Avanto) or TestTools/ Normal/ RX (Avanto Dot) tests should be used in case of problems in the RX-path. In case of the RCCS 10V error, refer to the special procedure described below.

Prerequisites

- MR system is on.
- Start the Service Software and select TestTools.

Performing the Procedures

TestTools

- **RCCS SGA Loop (High gain) (Functional)**

The stability and linearity of the SGAs are tested using the TTX signal of the modulator as the input signal to the 72 input channels of the RCCS.

Conclusion:

If this test fails, the corresponding SGA of the RCCS may be defective. (Also refer to TestTools/Online help).

- **RCCS SGA (TestTools Expert/ Normal/ RX/ RCCS SGA) (Avanto Dot Systems only)**

The stability and linearity of the SGAs are tested using the TTX signal of the modulator.

The TTX output signal of the modulator is fed to the RCCS. The RCCS input channels (with SGAs on high gain) are switched to the output in the following way:

Input channel 1-8, 9-16, 17-24, 25-32, 33-40, 41-48, 49-56, 57-64 to output channel 1-8 and the Body Coil channels 65-72 to output channel 1-8 once after another.

Conclusion:

If this test fails, the corresponding SGA of the RCCS may be defective. Also refer to TestTools/Online help and diagnosis of the test within TestTools.

- **RCCS Matrix Test ("Interactive") (Avanto and Espree Systems)**

The test requires about 2.5 hours (with maximum 32 outputs) and preferably should be performed overnight; the "Interactive" checkbox must be selected. Besides answering the pop-up in the very beginning of the test, no interaction is necessary.

The RCCS Matrix Test checks the 2048 +8 switch matrix nodes (64 inputs LC and 8 inputs BC, 8/12/24/32 outputs).

Conclusion:

If only single test measurements fail (signal too low), the RCCS is probably defective (switch node error).

If a complete row of adjacent nodal points fails (e.g. input #1 to input #64 to common output #5), the corresponding RX-cable connection (RCCS output #5 to receiver 1/#5) may be defective.

(Also refer to TestTools/Online help).

- **RCCS single switch (Interactive) (Avanto and Espree Systems)**

This is an interactive test with a user interface that allows you to select a combination of input channel and output channel of the RCCS. In addition it is possible to select a few fixed measurement times (quick, normal and long time stability).

(Also refer to TestTools/Online help).

- **RCCS plug1 - 10 (Interactive)**

The RF Service Plug must be connected to a coil connector 1 to 10 at the patient table.



When testing coil connector #1, the auxiliary cable has to be connected to coil connector #4 and the normal cable to coil connector #1.

When testing all other coil connectors, the auxiliary cable has to be connected to coil connector #1 and the normal cable to the coil connector under test.

The TX output signal of the modulator is fed via RCCS/multiplexer to coil plug 1 and via the auxiliary cable to the Service Plug/attenuating network. The signal goes back to the connector being tested and the cabling of the RCCS. Via the switching matrix, the signal is forwarded to receiver module # 1, input channel 1...8 based on the number of channels that are tested in the coil connector.

Conclusions:

In case of errors, the cabling from the coil connector (patient table) to the RCCS should be checked; refer to the tables of the signal lines below.

If the 10V preamplifier voltage is ok (refer to the **RCCS 10V Error** below), a preamplifier or switch matrix node of the RCCS may be defective.

RCCS 10V LC Preamplifier Supply Voltage Error

Troubleshooting Procedure

■ Identification of the error

The following **error messages** can be found in the Event Log:

Avanto Systems:

MRI_SUP_198: MRI_SUP_RCCSK2254_10VOLT_FOR_LC

MRI_SUP_195: MRI_SUP_RCCSK2254_SUM_VOLT or

Avanto Dot Systems:

MRI_RCC_1100 or MRI_RCC_1103

The scanner symbol in the status line is crossed out in red; no measurement is possible.

Click (with the left mouse key) on the error-id to get more information about the error and service action.

If one or more of the 64 voltage supplies for the LC amplifier are shorted or overloaded, the affected power supply unit is switched off permanently and the error is generated.

One or more of the 64 red error LEDs (channel 1 on top down to channel 64 at the bottom) at the right side of the RCCS light up (remove the magnet cover to get access). A permanent error will be latched in the RCCS and the corresponding LEDs light up permanently.

If the error occurs sporadically, it will be cleared after one second by the Software.

The following **warning message** can be found in the Event Log:

MRI_SUP_199: MRI_SUP_RCCSK2254_10VOLT_FOR_LC_WARNING

If the source for permanent error is removed, a "Scanner Reboot" is necessary. Alternatively, "percom" can be used to switch the 10V supply on again.



TestTools/Tools offers the function Autorecovery RCCS 10V Error enabled/ disabled (Avanto systems only). Refer to (→ TestTools/Tools / Page 88).



If all 64 LEDs are ON, another source for error, such as Software, communication, etc., is more likely. Check all supply voltages (delivered from RFIS/power supply and indicated by LEDs above connector RCCS/X1) and corresponding error messages concerning the RFIS.

■ Localization of possible error causes

Causes for the 10V error may be:

- Shorts in the coil connectors by conducting metal particles
- Damaged, distorted, or bent connector pins (LC or PTAB)
- Poorly soldered wires at the connectors
- Defective cables
- Defective LC preamplifier

- Defective RCCS

Use the tables below to localize the defective supply line.

■ Troubleshooting Actions

- Check all supply voltages of the RCCS
- Identify the affected supply line from the RCCS to the coil connector
- Disconnect the LC from the affected connector assembly and perform a **"RESET"**; see below. If the error cannot be reset, check the connectors for damaged pins or conducting particles. You may have to remove the coil connector assembly: Refer to (→ RF coil plugs / M6-020.841.15).
- Try to reproduce the error by repeatedly plugging the LC into the same and other coil plug assemblies and gently jiggling the plug.



To remove the LC plug, use the flap of the coil connector assembly as a lever.

- One after the other disconnect and visually check all corresponding cables and connectors in the affected channel from the RCCS until the error disappears after **"RESET"**.

"RESET":

- Perform "Reboot Scanner" or
- Start "percom" in a windows command shell:
- Select **RCCSK2254 (16) - Select Switch10Volt (22)**
 - » If the error cause has been removed, the preamplifier voltage will be switched ON again.
- Check the status: Select **get 10VoltStatus (23)**

Test of the 10 VDC Preamplifier Supply Voltage (with Service Plug)

The voltage supply for the internal RX preamplifier (+10V; 9.4 ...10.9 V) of the local coil can be checked and measured at the Service Plug, plugged into a coil connector. Measured with a DVM connected to the Service Plug sockets, values between 7.9V and 11.0V are ok.

The voltage must be continuously present independent of a running test. Green and/or yellow LEDs (according to the number of channels) indicate the status:

Green LED is on -> Voltages are within tolerance.

Only yellow LED's on -> Voltage is present but out of tolerance limits.

No LED on -> Cable from RCCS to coil connector may be shorted/open or RCCS is defective.

Check channel LEDs at RCCS; refer to RCCS 10V LC preamplifier supply voltage error. Refer to the description below.

The voltage can be tested with load on or off. Press the latch reset button on the Service Plug to start a new test.

Tables of Signal Lines and Connectors

Tab. 13 Signal lines between RCCS output and PTAB coil connector (1)

Coil connector -RX PIN		Connector RCCS	# RCCS LED
LC1 (8 ch.)	A1-A8	X311, 1-8	1-8
LC2 (4 ch.)	1,13,18,10	X311,9 and X312,1-3	9-12
LC3 (6 ch.)	A1-A6	X312,4-9	13-18
LC4 (8 ch.)	A1-A8	X321,1-8	19-26
LC5 (4 ch.)	1,13,18,10	X321,9 and X322,1-3	27-30
LC6 (6 ch.)	A1-A6	X322,4-9	31-36
LC7 (6 ch.)	A1-A6	X331,1-6	37-42
LC8 (8 ch.)	A1-A8	X331,7 and X332,1-7	43-50
LC9 (6 ch.)	A1-A6	X341,1-6	51-56
LC10 (8 ch.)	A1-A8	X341,7 and X342,1-7	57-64
TX signal PIN			
TX	PIN A9 on coil connector 1 (from TALES X6)		
TTX	PIN A1 on coil connector 1 (input from RCCS X60)		

Tab. 14 Signal lines between RCCS output and PTAB connectors (2)

RCCS connector	Cable	PTAB connect.	LC connect.
X311, X312	W5112	X102, foot X105, head	1-3
X321, X322	W5113	X103, foot X106, head	4-6
X331, X332	W5110	X101, foot	7,8
X341, X342	W5114	X104, foot	9,10

Fig. 4: Coil plug - top view (6/8 receive channels)

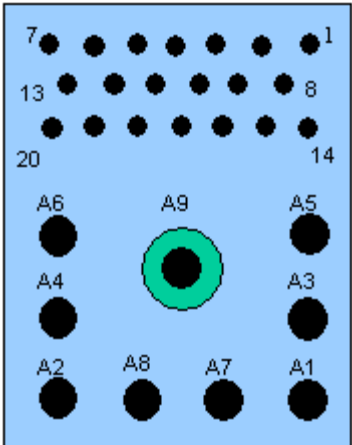
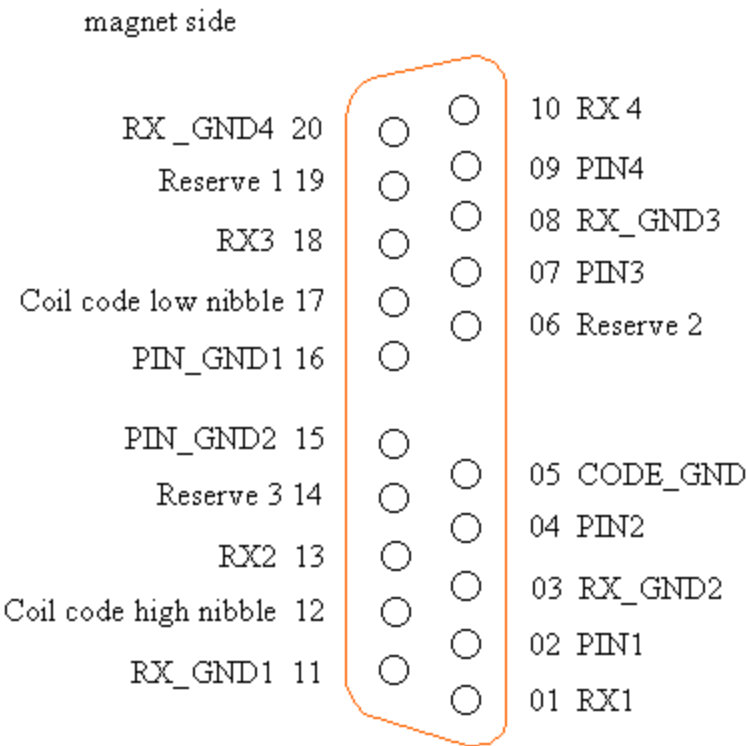


Fig. 5: Coil plug - 4 receive channels



2.4.4 RFIS Procedures

2.4.4.1 Prerequisites

- MR system is on.
- Start the Service Software and select TestTools/RF/RFIS or TestTools/ Interactive.
- Follow the instruction of the service procedure.

2.4.4.2 RFIS (Interactive) - Test PIN Diodes Control (LC) / DYSCON_16 (Avanto Systems)

i The Service Plug is used for these tests.

- Connect the Service Plug to the coil connector being tested; the auxiliary plug is not needed. The test can be run just once or in a loop (break with "Abort").
Avanto: TX-mode

Test Description

The 4, 6, or 8 green and yellow LEDs on the Service Plug (dependent on the number of channels of the coil plug) must light up as follows (with latch disabled):

Step 1:

Based on the number of coil connector channels, 4, 6 or 8 green LED's (status $U_R = -31V$ ok, static tuned) light up (idle state).

Step 2:

All green LEDs go out; yellow LEDs (status $I_F = 108\text{ mA}$ ok; static detuned) light up.

Step 3:

All yellow LEDs go out; green LEDs light up.

Step 4:

One after the other yellow LED lights up; green LEDs go out.

Step 5:

All yellow LEDs go out; green LEDs light up (idle state).

Example output of the test:

Fig. 6: Temporal sequence of tested coil plugs (2, 6, 8)

	Status	Tested PTAB Plugs
GetRFISTestResults	ok	2, 6, 8

LC Plug x (Avanto Dot Systems)

- Connect the Service Plug to the coil plug being tested; the auxiliary plug is not needed. The test can be run just once or in a loop (break with "Abort").
Avanto Dot: RX-mode.

Test Description

The 4, 6, or 8 green and yellow LEDs on the Service Plug (dependent on the number of channels of the coil plug) must light up as follows (with latch disabled):

Step 1:

Based on the number of coil connector channels, 4, 6 or 8 yellow LED's (status $U_R = -31V$ ok, static tuned) light up (idle state).

Step 2:

All yellow LEDs go out; green LEDs (status $I_F = 108\text{ mA}$ ok; static detuned) light up.

Step 3:

All green LEDs go out; yellow LEDs light up.

Step 4:

One after the other green LED lights up; yellow LEDs go out.

Step 5:

All green LEDs go out; yellow LEDs light up (idle state).

Conclusions

The test verifies the function of the RFIS/DYSCON_16 and the cables between RFIS and coil connector under test. If defective PIN diode channels are detected during the test (i.e. corresponding LEDs will not light up or switching between UR and IF is not ok), the DVM measurements described below will provide more detailed information regarding the possible cause for error.

Tab. 15 LEDs on Service Plug - LC PIN diode test

U_R green LED	I_F yellow LED	Conclusion
ON	ON	OK
OFF	ON	Error RFIS, if measurement is out of tolerance
ON	OFF	
OFF	OFF	Error RFIS or cabling, if voltage is out of tolerance

If the displayed sequence of tested coil plugs does not match the actual temporal sequence of tests, the following connectors at the RFIS could be interchanged:

RFIS X28 (coil connector 1, 2, 3) <-> RFIS X29 (coil connector 4, 5, 6)

RFIS X26 (coil connector 7, 8) <-> RFIS X27 (coil connector 9, 10)

2.4.4.3 RFIS PIN Diodes DVM Measurements (Interactive)

During the test, the PIN diode tune voltages U_R and detune current drop U_{measured} according detune current I_F can be measured with a DVM connected to the Service Plug sockets.

The I_F corresponding to the measured voltage can be calculated as:

$$I_F = 16.6 + 123.2 \times U_{\text{measured}}$$

- Reverse voltage (U_R) at PIN diode: -27.7...-31.3 VDC, typically - 29.8 VDC
- Voltage drop U_{measured} (I_F): + 0.65...+0.87 VDC, typically 0.74 V (corresponding to 108 mA).

To localize the defective DYSCON_16, swap the boards. If the error moves to another coil plug, the corresponding board is defective.

Measurement at RFIS

To differentiate between an RFIS or cable fault, also measure the voltages at connectors X26 - X29 at the RFIS:

-> Voltage mode: -31V

-> Current mode: typically +14V

Connectors at the RFIS

There are 4 modules DYSCON_16 (D1); the contribution of the modules via 4 connectors to the coil PIN diodes channels is shown in the table below.

Tab. 16 Description of RFIS connector

	X28 D14 Pin_sig (GND)	X29 D14 Pin_sig (GND)	X26 D14 Pin_sig (GND)	X27 D14 Pin_sig (GND)
to LC1	1-8 (9-12)			
to LC2	21-24 (25,26)			
to LC3	33-38 (39-41)			
to LC4		1-8 (9-12)		
to LC5		21-24 (25,26)		
to LC6		33-38 (39-41)		
to LC7			1-6 (9-11)	
to LC8			18-25 (26-29)	
to LC9				1-6 (9-11)
to LC10				18-25 (26-29)

Tab. 17 RFIS connectors to PAT Coil connectors

	from X28	from X29	from X26	from X27
# PAT LC Coil connector	1	4	7	9
Pin_sig	1 - 8	1 - 8	1 - 6	1 - 6
(GND)	(9 - 12)	(9 - 12)	(9 - 11)	(9 - 11)
	2	5	8	10
	2, 4, 7, 9	2, 4, 7, 9	1 - 8	1 - 8
	(16, 15)	(16, 15)	(9 - 12)	(9 - 12)
	3	6		
	1 - 6	1 - 6		
	(9 - 11)	(9 - 11)		

Tab. 18 Contribution DYSCON_16 module to coil connector

DYSCON_16 (left->right)	Module 1	Module 2	Module 3	Module 3
Coil connector	1 (1-8)	3 (5,6)	6 (3-6)	8 (7,8)
(channel)	2 (1-4)	4 (1-8)	7 (1-6)	9 (1-6)
	3 (1-4)	5 (1-4)	8 (1-6)	10 (1-8)
		6 (1-2)		

2.4.4.4 Body Coil Test/ PIN Diodes/ DYSCON_BC_5A



This test "BC Check Tuning" is a subtest of TestTools/RF/BC.



This test is also performed automatically on every "Reboot scanner".

Fig. 7: BC test result (RFIS PIN diodes test)

BC tuning test results

	Value
Tune current	ok
Detune voltage	ok
Static 500V	ok

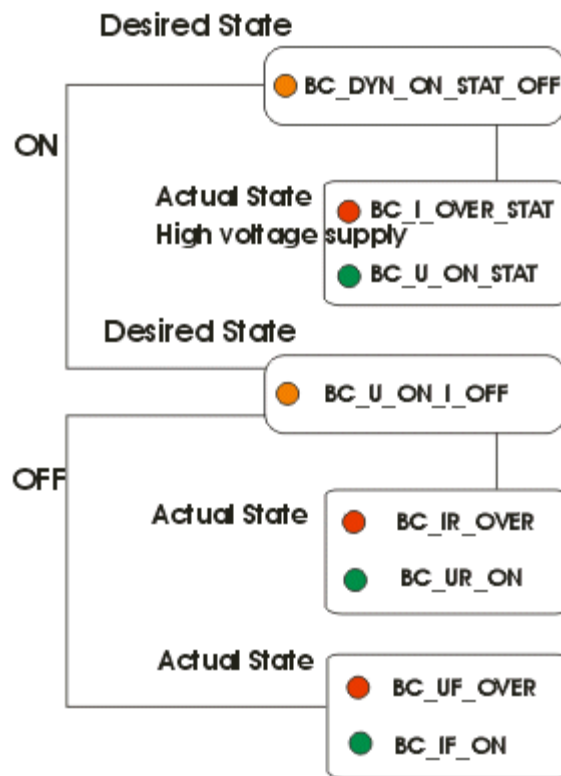
The test **BC Check Tuning** checks the BC Tuning/PIN diodes control/ **DYSCON_BC_5A** against specification.

Tune current: +15V/5000 mA +/- 325 mA

Dynamic detune voltage: -29.8V -1.5V/ +2.1V

Static detune voltage: -360V ... -460V

Fig. 8: BC Test - LED display BC Mode on RFIS



The status of the DYSCON_BC_5Acan is displayed by LEDs at the RFIS.

For the description of the LEDs, also refer to the Functional Description/RF System/RFIS.

Description of BC Mode LEDs

Tab. 19 BC mode LEDs at RFIS

LED	ON	OFF
BC_DYN_ON_STAT_OFF Yellow	Dynamic mode	Static mode (-450V)
BC_U_ON_I_OFF Yellow	dyn.detuned (-31V)	dyn. tuned (+15V/5A)
BC_I_OVER_STAT Red	Overcurrent in -450V source: defective DYSCON_BC_5A, PIN diodes of BC or short in cable	If BC_U_ON_STAT is on: -450 V ok If BC_U_ON_STAT is off: -450V source defective; voltage level is too low Check 48 ACV supply Defective DYSCON_BC_5A
BC_U_ON_STAT Green	-450 V ok	-450 V source level defective; voltage level is too low Check 48 ACV supply Defective DYSCON_BC_5A

LED	ON	OFF
BC_IR_OVER Red	Overcurrent in dynamic -31V source defective DYS-CON_BC_5A, PIN diodes of BC or short in cable	If BC_UR_ON is on: -31V ok If BC_UR_ON is off: -31 V source level too low or supervision on DYS-CON_BC_5A defective
BC_UR_ON Green	Dynamic detune voltage -31 V ok	-31 V source level too low or supervision on DYS-CON_BC_5A defective
BC_UF_OVER Red	Dynamic current source at upper voltage limit: Broken cable?	Dynamic current source +15V/5000mA ok
BC_IF_ON Green	Dynamic current level ok	Dynamic current level too low or supervision on DYS-CON_BC_5A defective

If the test fails, check the **Event Log** for RFIS error messages. Double-click the **Error ID** to get more information about the error and service actions.

Typical error messages:

Avanto Systems:

MRI_COH_840: MRI_COH_COH_ERROR_TEST_BC_STATUS

MRI_COH_841: MRI_COH_COH_ERROR_TEST_BC_STATUS1

.....

MRI_COH_846: MRI_COH_COH_ERROR_TEST_BC_STATUS6

Avanto Dot Systems:

MRI_RIS_840

MRI_RIS_1005

Conclusions

If the test fails, the error could be caused by:

- DYSCON_BC_5A
- Fuse F3
- Low RFIS supply voltages (check corresponding error messages)
- Missing 48 ACV supply from PTAB (connector RFIS/X6)
- Short in cabling to BC
- Defective PIN diodes of BC

Quick check of the PIN diodes of the Body Coil:

To check the PIN diodes, remove the connector X3 of the detuning unit at the BC and measure using the "diode test function" of a DVM in both (plus/minus) directions at the detuning unit.

Typical values shown: Forward direction: 0.4 V
Reverse direction: "OL"

TestTools/ Normal/ TX/ BCTune/ Detune (Avanto Dot Systems only)

With this test the tune/detune circuit of the Body Coil including the RFIS is tested:

- tuned: BC Parameters for tuned state (current from RFIS)
- detuned: BC Parameters for detuned state (voltage from RFIS)

The test switches the PIN Diodes signals from the RFIS to the BC to the 3 possible states:

- current
- low voltage
- high voltage

The RFIS is checked for an error status that occurs when one of these states cannot be established due to a shortcut or a broken line.

Description of Status LEDs on RFIS

- Check LEDs on the RFIS, if RFIS-related error messages occur.

Tab. 20 Status LEDs at RFIS

LED	ON	OFF
RESET Yellow	Reset all memories and both COSLIOS by pressing the RESET button	Idle state
SETTING_SELECT 1-4 yellow	LC/BC Setting selection	
DYN_CTRL Yellow	Dynamic switching of PIN Diodes of LC and BC	
N_SEQUENCE_ON Yellow	Sequence off; all channels in dynamic voltage mode (-31V)	Sequence on; status of channels determined by DYN_CTRL, setting and configuration of selected setting
PICKUP_SELECT Yellow	Pick-up coil selected	Pick-up coil deselected
RESERVE Yellow	Not used	
CAN_TX1 Yellow	CAN Bus TX1 active (LED flashing)	CAN Bus TX1 deactivated
CAN_TX1 Yellow	CAN Bus TX2 active (LED flashing)	CAN Bus TX2 deactivated

LED	ON	OFF
CAN_RX1 Yellow	CAN Bus RX1 active (LED flashing)	CAN Bus RX1 deactivated
CAN_RX2 Yellow	CAN Bus RX2 active (LED flashing)	CAN Bus RX 2 deactivated
CLOCK_DISABLED Yellow	Clock for FPGA is disabled, e.g. sleep mode, no communication between COSLIOS and FPGA	Clock for FPGA enabled, awake state, communication between COSLIOS and FPGA possible
VOLT_OK Green	All supervised supply voltages are OK	At least one of the supply voltages is missing
CONFIG_DONE Green	Configuration of FPGA ok; RFIS ready.	Configuration of FPGA not yet finished; RFIS not ready.
CODE_INTERRUPT Red	Error LED lights up briefly if a coil/ code change occurs	No coil change

Yellow LEDs LC x_PIN y UR_ON:

ON: U_R (-31V) on; (voltage mode)

OFF: I_F (108 mA) on; (current mode)

2.5 Gradient/Body Coil (GC/BC) Espree

2.5.1 Safety Hints



WARNING

During measurements at the BC rods, you have to work closely to the non-isolated connections of the gradient cables at the gradient coil.

Hazardous voltages may be present at the connections of the gradient cables.

- » Switch off the gradient power amplifier via circuit breaker F2/GSSU and F1/100 A contactor in the GPA. Wait at least 20 minutes before you start any action at the BC.



WARNING

Life-threatening voltages may be present on the BC rods (DCV -450V).

- » Carefully check the status of the RFIS output - indicated by LEDs at the RFIS - before working at the BC rods.

2.5.2 BC Problems - Possibly Affected Components

The following list shows the components that need to be checked since they may be responsible for BC problems.

- BCCS
- TALES
- RFIS power supply unit
- 48 ACV voltage supply from PTAB (connector RFIS/X6)
- RFIS:
 - DYSCON_BC_5A
 - Fuse F3
 - Cable between RFIS (X9) and BC (short/ open)
 - BC rods:
 - PIN diodes (short/ open)
 - Inductors (open)
 - Capacitors (broken/ burnt down)
 - Broken solder connections (from rods to RF shield)

The following table helps to identify the defective component. (→ Tab. 21 Page 44)

2.5.3 General RF - Troubleshooting

Tab. 21 RF-troubleshooting

Step	Status/Symptom	Possible Cause	Service Action	Reference
1.	Low S/N of local coils RFIS error messages after booting the system: Error BC status test 1/2/3.	- LC defective - RFIS power supply units defective - RFIS DYSCON_BC_5A defective - Shorted PIN-diodes BC or cable BC to RFIS	- Perform QA/ Coil Check - Try another LC (same type if available) - Perform RFIS procedure. - Perform "Procedure before removing the bore cover".	RFIS procedures: (→ Body Coil Test/ PIN Diodes/ DYS-CON_BC_5A / Page 38) (→ Procedure before Removing the Bore Cover / Page 46)

Step	Status/Symptom	Possible Cause	Service Action	Reference
2.	Increased ADJ/TRA voltage SAR problems during normal operation High RF-reflection	Problem of components in the Tx-path	- Check manual ADJ/TRA with loaded large spherical: Typical ok-value < 450 V; if increasing, go to step 2.2. - Run Tune-up/ Tuning Calibration; if it fails, go to step 2.1 - Run Tune-up/ RF Characteristic	Error messages of Tune-up functions Error diagnosis of TestTools
2.1		- Defective BCCS - Defective TALEs - Defective RF-cables	- Run TestTools/ BCCS/RCCS (functional , interactive) - Run TestTools/ RFAS - Repeat the TALEs calibration and use the BCCS outputs instead of the TALEs outputs; if ok, replace TALEs and/or RF-cables BCCS-TALEs.	TestTools Online Help TestTools diagnosis
2.2		Growing BC problem	Run TestTools/ BC. Compare the result with previous test results that were run during the installation of the system.	(→ Procedure before Removing the Bore Cover / Page 46)

2.5.4 Procedure before Removing the Bore Cover

- If error messages from RFIS appear after booting (failing BC-Test): Open Seso/ Eventlog and click the "ID" (blue number) to get additional information for service actions.
- Check Error LEDs RFIS: (→ Body Coil Test/ PIN Diodes/ DYSCON_BC_5A / Page 38)
 - Check RFIS power supply unit
 - Check 48 ACV power supply unit from PTAB (RFIS/ X6)
- If error in BC status test 2/3:

Remove plug X9 from RFIS ; if the green LED "BC_UR_ON" lights up -> short of PIN diodes or in cable <W5103>.

Measure with DVM/Diode test on plug X9:

Forward direction approximately 0.4 V

Reverse direction: "OL".
- If error in BC status test 1/2/3:

Check fuse RFIS/ F3. Replace DYSCON_BC_5A if there is no 5A current flow in case that cable RFIS/BC is not broken and supply voltages of RFIS are ok.
- **Run TestTools/BC**

Compare the results also with the measurement taken during the installation of the system.

Specifications:

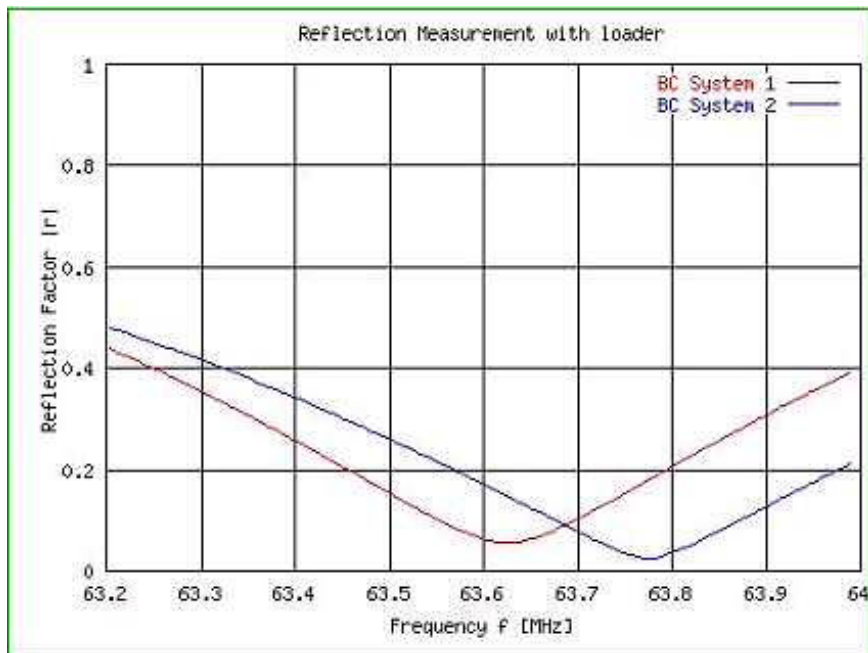
 - 1) The minimum reflection should be less than 0.3 for both systems.
 - 2) The frequency, where minimum reflection occurs, must be between 63.45 MHz and 63.85 MHz.
 - 3) The 2 frequencies, where minimum reflection occurs, should not differ by more than 100 kHz.
 - 4) The maximum transmission should be lower than -14 dB.

Example:

The figure (→ Fig. 9 Page 47) shows relatively good results with a reflection factor at both systems that is lower than 0.2 at 63.6 MHz.

The difference frequency of the minimum should normally be lower than 100 kHz. In the example the difference is 140 kHz.

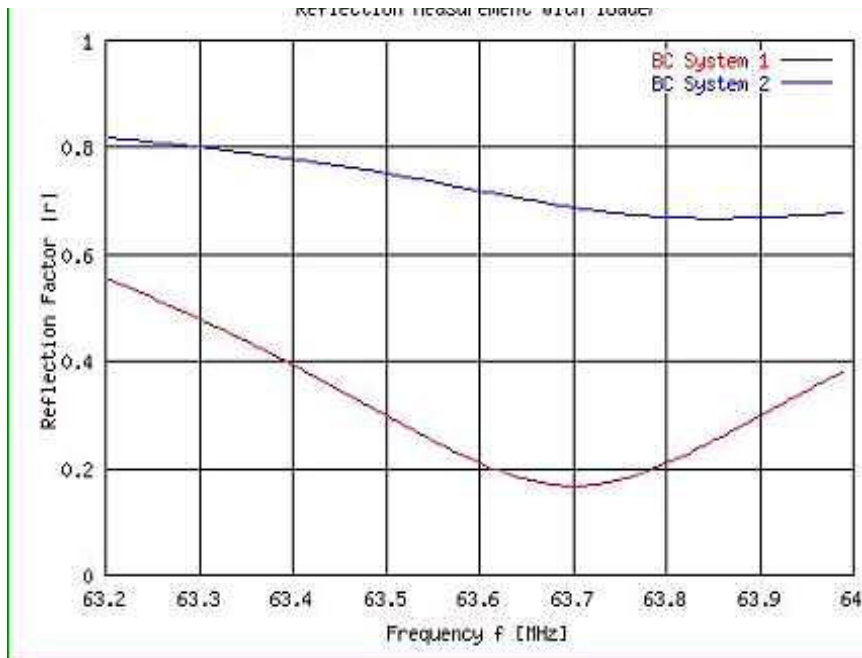
Fig. 9: TestTools/BC - result is ok



The figure (→ Fig. 10 Page 47) shows a result which is not acceptable. Both minima are shifted to higher frequencies and system two has a quite high reflection.

The measurement was performed on the same system a few days later and clearly shows the variation in time. This shows that something is going wrong with the BC.

Fig. 10: TestTools/BC - result is not ok



- Contact the HSC/MR group in Erlangen for a review of previous service actions/measurements and for a decision regarding further actions.

2.5.5 Procedure after Removing the Bore Cover

2.5.5.1 First Checks:

- Check the surface of the bore cover for burns from damaged capacitors.
- Visually check the BC rods for damage.

Hint:

The rods below and above belong to system 1, while the rods at the right and left side belong to system 2.

Other rods have influences on both systems.

2.5.5.2 Voltage Measurements (with DMM)

Preliminary Step:

- System is running: Pull the FOC U5/U6 at RFIS.
 - » 5A (IF) is running, RFIS green LED (BC_IF_ON) is ON.

or:

- System is in error condition:
 - start "percom" (→ Percom / Page 58)
 - choose item **1**. "RFISK2254"
 - choose item **33**. "Set BC mode for all settings"
 - Choose "static current 5A"
 - » 5A (IF) is running, RFIS green LED (BC_IF_ON) is ON.

For error analysis and description of BC mode LEDs: (→ Body Coil Test/ PIN Diodes/ DYS-CON_BC_5A / Page 38)

DVM - measurements:

- Measure with a DVM at the points shown in Fig x/y of all 16 BC rods according to the table (→ Tab. 22 Page 51). TP PIN D is equivalent to the PIN diodes supply ground (stripe from cable connector to the rods).



Use long measuring cables to keep the magnetic DVM far enough from the magnet.
A second person should read the DVM display.

The rods of the BC are accessible through the E-field shield and contain capacitors, inductors, and PIN diodes for the detuning.

Fig. 11: Schematics of BC rod (PIN power supply hidden below E-field shield)

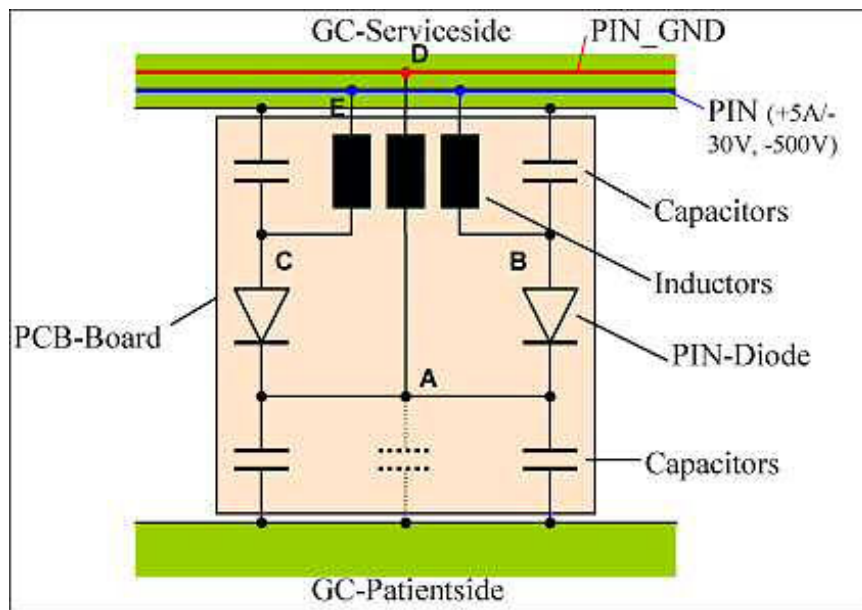
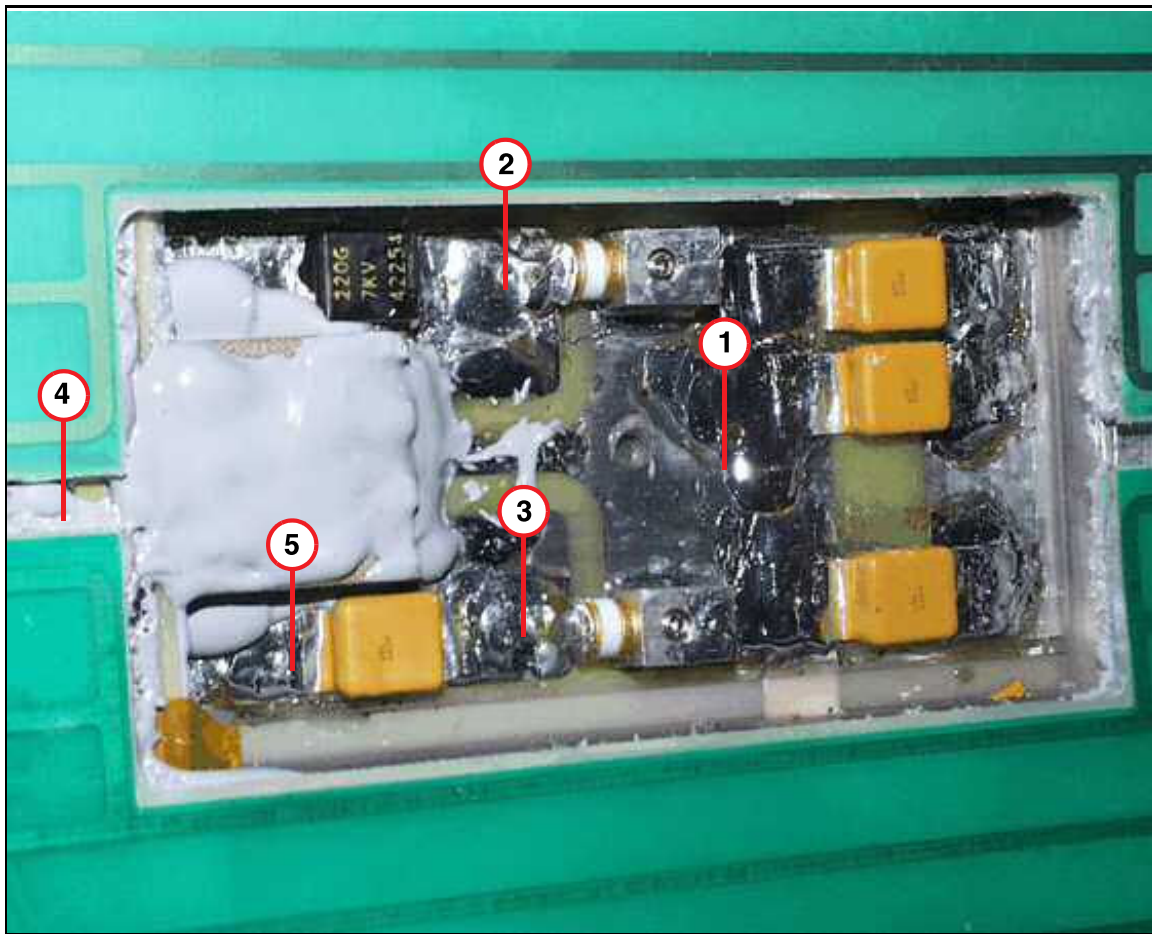
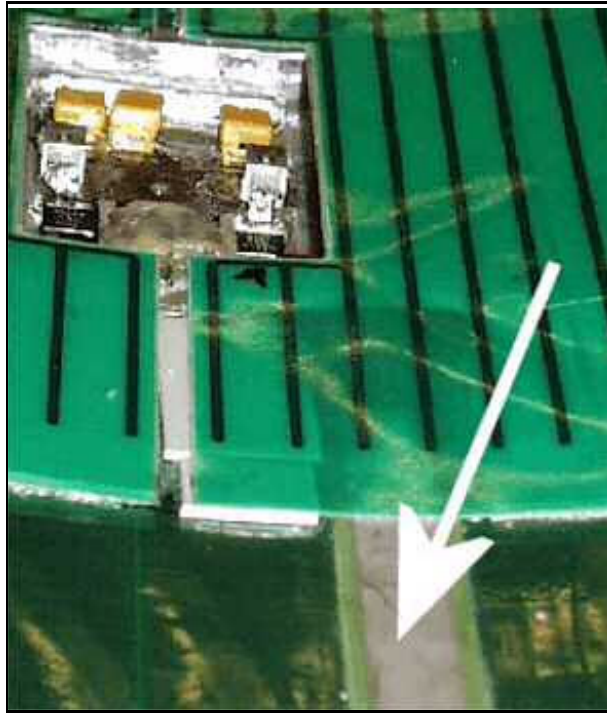


Fig. 12: Detailed view of BC rod with test points (TP)



- (1) TP A
- (2) TP B
- (3) TP C
- (4) TP D PIN_GND
- (5) TP E

Fig. 13: Testpoint D PIN_GND



- First measure all 16 rods at test point (A - D) and compare with the values in the table (→ Tab. 22 Page 51).
You can use the PIN_GND indicated in (→ Fig. 13 Page 51).
- Contact HSC/Erlangen for further action.

Tab. 22 DVM measurements at BC rods

	Voltages on defective rods test points (if only one rod is involved)				Other rods
Error	D - A (V)	D - B (V)	D - C (V)	D - E (V)	D - A (V)
All OK	0.64	1.4	1.4	1.7	0.64
Open PIN-D (right)	0.5	1.3	1.7	1.7	0.64
Open PIN-D (left)	0.5	1.7	1.3	1.7	0.64
Open PIN-D (both)	0.05	1.75	1.75	1.75	0.68
Short PIN-D (right) *)	0.9	1.6	0.9	1.7	0.64
Short PIN-D (left) *)	0.9	0.9	1.6	1.7	0.64
Short PIN-D (both)	1.1	1.1	1.1	1.65	0.62
*) After system reboot: Error BC Test 2 and Test 3					
Open inductor (middle)	1.7	1.78	1.78	1.78	0.7
Open inductor (right)	0.48	1.28	0.4	1.73	0.62
Open inductor (left)	0.48	0.4	1.28	1.73	0.62

	Voltages on defective rods test points (if only one rod is involved)				Other rods
Open inductor (both)	0.05	0.04	0.04	1.78	0.7
If more than one component is defective, the overall voltage levels will change. Contact HSC/MR in Erlangen for evaluation.					

There are now 2 possibilities:

- **Repair of GC/BC** by replacing shorted or open PIN diodes: Contact HSC/Erlangen for support.
- **Replacement of GC/BC** in case of defective capacitors, inductors or open solder connections (→ Upgrade Espree straight bore Gradient/Body coil to Slim-line / M6-020.841).

2.6 Additional Procedures (Tune-up)

2.6.1 Tuning Calibration - Tune-up - Expert Mode

The tuning steps can be selected separately.

Fig. 14: Tuning calibration (Expert mode)

TuneUp Expert - Tuning Calibration				
Tales		X4		
		open	50 Ohm	short
X3	open			
	50 Ohm			
	short			

- » After successful completion of TALES calibration, the BC tests can be used.

2.6.1.1 Common Problems

- **External cable connections are not OK:**
 - **Check Transmit path:** Modulator X13/TTX - ACC top X42
 - <W0185> - RF-filter plate X35A - <W1185> - RCCS X60
 - **Check Receive path:** Receiver/MR SIG 1-32 - CCA top X1 to X32 - RF-filter plate X01A to X32A - RCCS X501 to X532 cables: MRSIG 1-8 <W018x> between ACC and RF-filter plate (x = 1...4) <W118x> between RF-filter plate and RCCS (x = 1...4)
- **Bad 50-Ohm Terminators:**
 - Use a DVM to verify.
- **Wrong termination during the calibration:**
 - 50 Ohms - short - open

2.6.1.2 Error Messages - Recommended Tests

If the error message indicates TALES... the TALES itself may not necessarily be the problem. It can be any one of the items mentioned above. Certain tests are necessary to isolate the problem.

TALES-related Messages

"TALES calibration data is not meaningful."

"TALES verification is out of specification."

- » Check cables
- » Check if the terminators used are OK. Replace TALES if not successful.

2.6.2 RF Chara - Tune-up - Expert Mode

Fig. 15: RF Characteristic (Expert mode - workflow popupwindow)

RFPA		RFPAK2204	
Path		BC	LC
Nucleus	1H	☒	☐
	3He	☐	☐
	19F	☐	☐

RF Maximum Voltage Volt

Settings

☐ No adjustments

☐ No switch to dummy

OK Cancel

Several measurement options can be selected:

- Maximum RF voltage
- Selection of RFPA (If spectroscopy option is installed)
- Selection of TX-path (BC/LC)
- Selection of TX-frequency/nucleus (If spectroscopy option is installed)
- Measurement with no adjustments (frequency, ...)
- Disabling of automatic switching to dummy (TAS_3TM)

2.6.2.1 Possible RF Chara measurements in case of Problems

- Select a lower maximum voltage.
- Perform the RF characteristic measurement without adjustments (especially frequency adjustments) while using the TAS_3TM dummy load or an external 50 Ohm power dummy for the measurement.

(The frequency can be manually typed in: Options/Adjustments).

2.6.2.2 Error Messages - Recommended Tests

Frequency Adjustment:

"Frequency out of range"

The magnet current has to be adjusted (ramped up). Use the frequency from the shim result and calculate the new current setting: $J_{\text{new}} = J_{\text{actual}} \times (f_{\text{upper limit}}/f_{\text{actual}})$

for AVANTO /ESPREE $f_{\text{upper limit}} = 63.7 \text{ MHz}$
or 1.5 T

(Tolerance range $f_{\text{upper limit}}: +0/-5\text{KHz}$)

"Signal too low"

- Check if the cabling between BCCS (X9/X3) and RCCS (X41/42) is reversed.
- Phantom not centered: Reposition the phantom
- Field homogeneity out of spec: Reshim the magnet
- Receive path not ok: Use another coil for test purposes only
- Perform the BC and the RCCS/plug x (x = 1 to 10) tests

Calibration Voltage Search:

"Measured voltage is less than "min_voltage"

(min_voltage = 10V)

"Error in expected RF output: Check RF amplifier"

The actual voltage level at the TALES is less than 10V, while the nominal start value is already 100V.

- Perform TestTools/RF/RFAS test. Most likely, there is a problem with the TALES.

"Trying to set voltage to TRA voltage"

"Cannot set to value greater than max_lim_voltage"

(max_lim_voltage is 800V)

"Error in expected RF output: Check RF amplifier"

The system tries to find the nominal voltage, i.e. 566V at the TALES.

This error occurs if the calculated /nominal value is > 800V in order to get the correct TALES value.

- Perform TestTools/RF/RFAS test. Most likely, there is a problem with the TALES.

"Number of allowed search measurements exceeded"**"Error in reaching or finding calibration voltage"**

The search measurement makes 25 attempts to find the 566V levels before the above error is generated.

There could be an RF instability problem in either the transmit or the receive path.

- Perform TestTools/RF tests.

RF Calibration:

If any of the following errors occur, it is necessary to check the RF power using a URV5.

Most likely, there is a problem with the RFPA DICO.

- Perform TestTools/RF/RFPA/RFAS test.

"Value of PF ADC conversion is too high"**"Please check the RF components, especially the RFPA DICO"**

The DICO factor will be calculated using the "voltage search" results:

Dico factor = TALESVoltage/DICO ADC; it must be < 0.2.

"ADC value of ADC is greater than maximum"**"Calibration is too high: Please check RF components, especially RFPA DICO")**

The expected ADC value is out of range; ADC overflow.

"Value of peak power is too high"

There is a difference between the nominal value and the value measured with the DICO.

Most likely, there is a problem with the DICO (stability or linearity).

RF Max Power Search:**"Measured voltage is less than "min_Voltage"**

(min_voltage = 10V)

"Error in expected RF output; check RF amplifier"

One of the involved components failed just now; most likely the RFPA.

- Perform Testtools/RF/RFPA tests.

"Trying to set voltage to "TRA-Voltage"**"Cannot set to value greater than max. linear voltage"****"Error in expected RF output! Check RF components"**

The RFPA does not reach specifications. The reason could be:

The level of the SSB-signal (TX-Module) or the RFPA/DICO itself is faulty.

- Perform TestTools/RF/MOD/REC test.

"Number of allowed search measurements exceeded"**"Error in reaching or finding maximum power"****"Please check RF amplifier and RF components"**

It was possible to reach the max. power with the previous measurement. There could be an instability problem in the transmitter or even in the receiver path.

- Perform TestTools/RF tests.

2.7 Procedures for Special Problems

The following presents test procedures for special problems (e.g. service actions following error messages of RFSWD or related to RF power).

2.7.1 Percom

This tool allows direct control of components, e.g. switching the TX-path in the TAS (BC/LC/Dummy).



WARNING

When using "percom", it is possible to set the MR system to a mode where patient safety modules are effectively disabled.

Measuring with these safety modules disabled may lead to permanent injuries. It is also possible to measure using an incorrect image position.

- » Only specially trained persons with knowledge of these risks are permitted to use "percom".

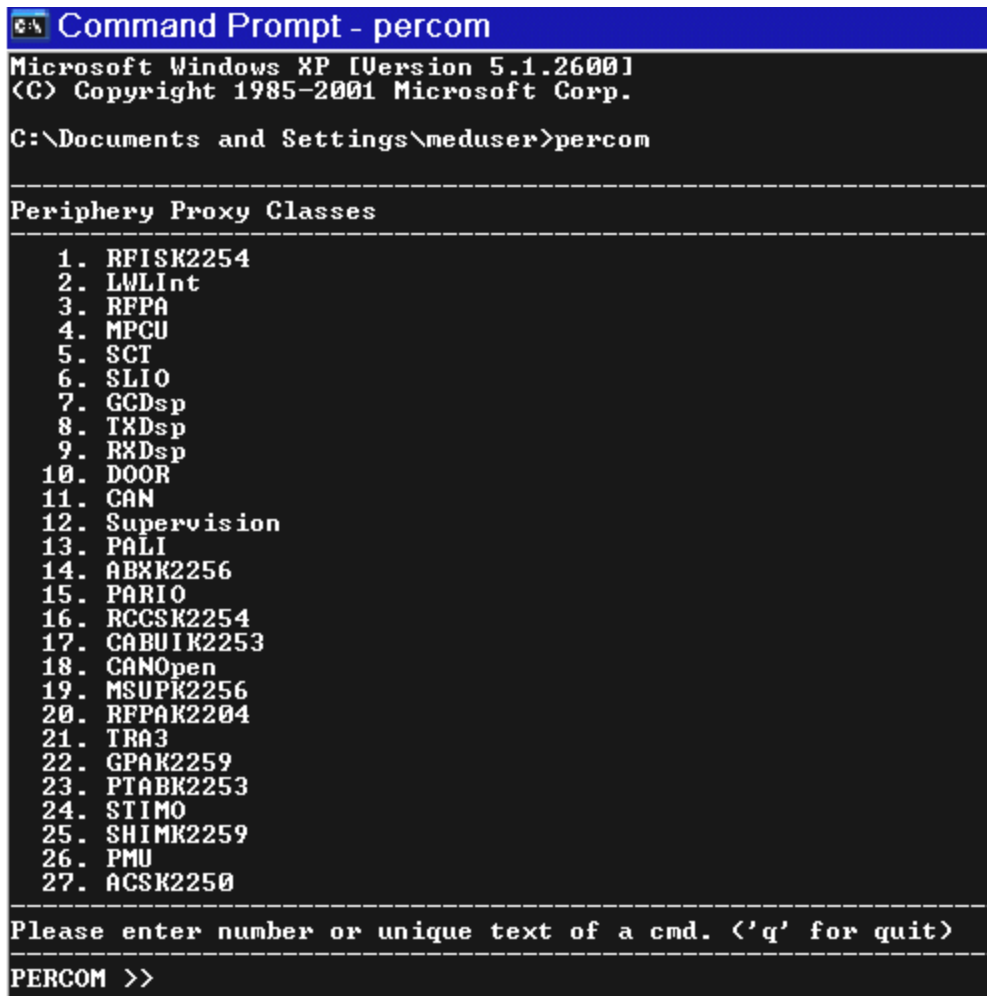
You need a password for using "percom".

2.7.1.1 Creating the Password

- When starting the tool (typing "percom" in windows command shell) for the first time on a system, type in a "?" when you are prompted for the password.
- A key such as 'as95np/o' will be returned.
- Call the HSC/MR group in Erlangen for creating the password.
- After restarting the tool, type in the specific password.

Example of using Percom

Fig. 16: percom start menu



```
Command Prompt - percom
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\meduser>percom

-----
Periphery Proxy Classes
-----
 1. RFISK2254
 2. LWLIInt
 3. RFPa
 4. MPCU
 5. SCT
 6. SLIO
 7. GCDsp
 8. TXDsp
 9. RXDsp
10. DOOR
11. CAN
12. Supervision
13. PALI
14. ABXK2256
15. PARIO
16. RCCSK2254
17. CABUIK2253
18. CANOpen
19. MSUPK2256
20. RFPaK2204
21. TRA3
22. GPaK2259
23. PTABK2253
24. STIMO
25. SHIMK2259
26. PMU
27. ACSIK2250

-----
Please enter number or unique text of a cmd. <'q' for quit>
PERCOM >>
```

Fig. 17: Selection menu for RCCS functions

```

Command Prompt - percom
25. SHIMK2259
26. PMU
27. ACSI2250

Please enter number or unique text
PERCOM >>16

Member functions of <RCCSK2254>
1. dump
2. setTrace
3. setVirtualDevice
4. getStatus
5. printStatus
6. getLoadwareName
7. sleep
8. awake
9. restart
10. setControllerState
11. getHWInfo
12. getParameter
13. setDefaultValue
14. setParameter
15. storeParameters
16. restoreParameters
17. getWakeupCounter
18. resetWakeupCounter
19. setCOHTraceLevel
20. getCOHTraceLevel
21. configure
22. switch10Volt
23. get10VoltStatus
24. setNoOfReceiverChannels

```

- Select the component and action by typing in the appropriate number in the percom menus.

2.7.2 Readout and Check of the Settings of RCCS/ RFIS using the fMOB-Test Tool

(fMOB = Meas Object Browser - TestTool)

2.7.2.1 Case of Application

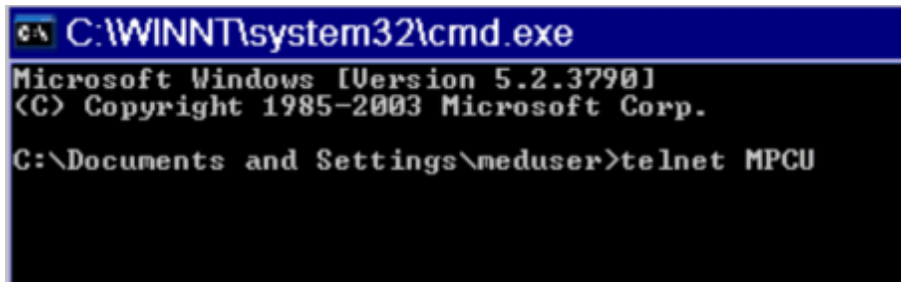
Possible defective FOCs e.g. for the dynamic control signals SEL_ 0-3 to RCCS or RFIS can be found using the test tool fMOB.

In case of sporadic errors, i.e. loss of signal of local coil elements, it is useful to readout the PIN-diode desired settings at the RCCS or RFIS using fMOB and to compare this with the actual setting (shown by LEDs at the component) after running a measurement.

2.7.2.2 Starting fMOB

- Open a cmd window and type: telnet mpcu.

Fig. 18: Starting fMOB



2.7.2.3 Selecting the RCCS/ RFIS Settings

- In the fMOB tool select the following menu items:
 - 2 Measurement
 - 8 libSSL
 - 1 SSLCtrl
 - 10 RCCSsettings or - 11 RFISsettings

Fig. 19: fMOB menus

```

Telnet mpcu
-> fMOB

-----
1. MeasPALImpl          1
2. Measurement          348/355
3. PMUMeasCtrlImpl     1
4. SSLCtrlPalIPPLimit  1
5. Seq                  6
6. libEQ                11
7. libIDUMP             1
8. libSC                 9
-----
Please enter number or unique text of a cmd. <'q' for quit>
MOB> 2

Measurement
-----
1. MeasSections          4
2. MeasUtils             61
3. Seq                   2
4. libEQ                109/114
5. libIQ                 3
6. libRT                 3
7. libSC                164/166
8. libSSL                2
-----
Please enter number or unique text of a cmd. <'q' for quit>
MOB> 8

Measurement/libSSL
-----
1. SSLCtrl              SSLCtrl
2. XPulse               XPulse
-----
Please enter number or unique text of a cmd. <'q' for quit>
MOB> 1
Nothing to dump

Methods of Object 'SSLCtrl'
-----
1. dump
2. debug
3. reset
4. edit
5. Periphery
6. RFChara
7. GCSPara
8. TXAdjP0
9. PalIPPLimit
10. RCCSsettings
11. RFISsettings
12. AdjData
13. AdjMDSResult
-----
Please enter number or unique text of a cmd. <'q' for quit>
Cmd >

```

- Type 'q' for quit.

2.7.2.4 Example for readout of RCCS Setting (BC Measurement)

- Run a measurement with the Body coil, e.g. a "haste" sequence.
- Check the desired setting of RCCS using fMOB:

Fig. 20: RCCS Setting: 2

```

Methods of Object 'SSLCtrl'
-----
1. dump
2. debug
3. reset
4. edit
5. Periphery
6. RFChara
7. GCSitePara
8. TXAdjF0
9. PaliPPLimit
10. RCCSsettings
11. RFISettings
12. AdjData
13. AdjMDSResult
-----
Please enter number or unique text of a cmd. ('q' for quit)
Cmd > 10
-----
Dump of the RCCS settings
-----
Setting: 2
-----
CoiID : ElementID : Coil Plug : MUX channel : -> : RxChannel :      Gain
!      !          !          !          ! -> !      ! dynamic <CWBit set -> low>
-----
TTXPath = RF TUNE OFF

```

- Check LEDs at RCCS: LED SEL₁ is on -> corresponds to 2¹ = Setting 2, (all ok).

Tab. 23 Setting of RCCS/ RFIS (LED display) dependent on SEL₀₋₃ signals

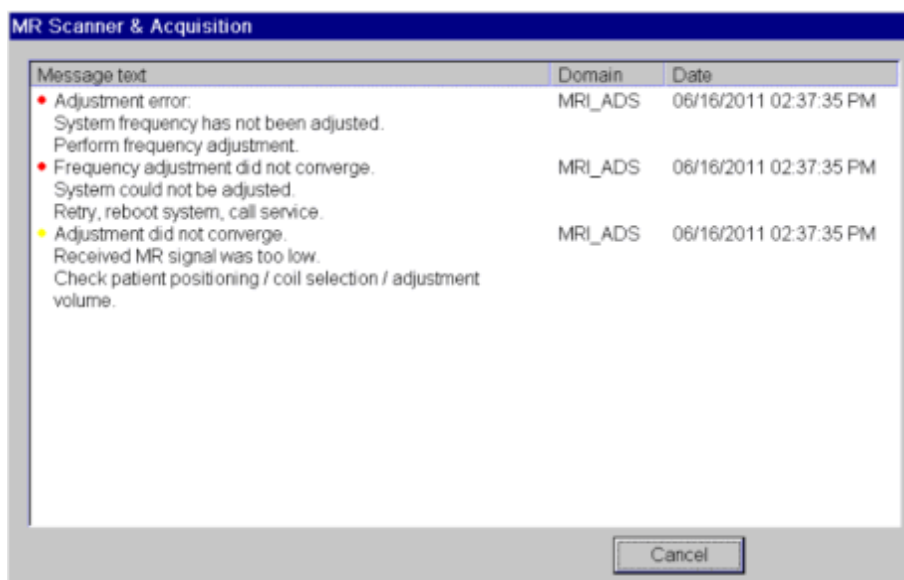
Setting	SEL ₀	SEL ₁	SEL ₂	SEL ₃
0	off	off	off	off
1	on	off	off	off
2	off	on	off	off
3	on	on	off	off
4	off	off	on	off
5	on	off	on	off
6	off	on	on	off
7	on	on	on	off
8	off	off	off	on
9	on	off	off	on
10	off	on	off	on
11	on	on	off	on
12	off	off	on	on

Setting	SEL_0	SEL_1	SEL_2	SEL_3
13	on	off	on	on
14	off	on	on	on
15	on	on	on	on

2.7.2.5 Example for Troubleshooting with the fMOB Tool

- Use the Head Matrix coil for the measurement. (Use correct phantom and center the coil in the magnet).
- Assuming the RFIS_SEL1 signal is broken. (Simulate by pulling the LWL_INT/U16).
- Run manual adjustment ADJ_FREQ.
 - > The measurement fails, result is only noise; an error message will come up:

Fig. 21: Error messages



- Check the desired setting of RFIS using fMOB:

Fig. 22: RFIS Setting: 11

```

Methods of Object 'SSLCtrl'
-----
1. dump
2. debug
3. reset
4. edit
5. Periphery
6. RFChara
7. GCSitePara
8. TXAdjF0
9. PaliPPLimit
10. RCCSsettings
11. RFISsettings
12. AdjData
13. AdjMDSResult
-----
Please enter number or unique text of a cmd. (<'q' for quit>)
Cnd > 11
-----
Dump of the RFIS settings
-----
Setting: 11
-----
CoilID      : ElementID : Coil Plug : PIN channel : PIN mode
-----
HeadMatrix  : HE2       : LC3       : 13          : current (dynamic)
HeadMatrix  : H2S       : LC3       : 14          : current (dynamic)
HeadMatrix  : H2T       : LC3       : 15          : current (dynamic)
HeadMatrix  : HE4       : LC3       : 16          : current (dynamic)
HeadMatrix  : H4S       : LC3       : 17          : current (dynamic)
HeadMatrix  : H4T       : LC3       : 18          : current (dynamic)
HeadMatrix  : HE1       : LC6       : 31          : current (dynamic)
HeadMatrix  : H1S       : LC6       : 32          : current (dynamic)
HeadMatrix  : H1T       : LC6       : 33          : current (dynamic)
HeadMatrix  : HE3       : LC6       : 34          : current (dynamic)
HeadMatrix  : H3S       : LC6       : 35          : current (dynamic)
HeadMatrix  : H3T       : LC6       : 36          : current (dynamic)
Body        : BC        : BC        : 65          : current (dynamic)

```

- Check LEDs at right front side of RFIS: LEDs SETTING_SELECT 0 and 3 are on
 -> corresponds to $2^0 + 2^3 = 9$ -> actual Setting 9 (not ok, the correct setting should be $11 = 2^0 + 2^1 + 2^3$).
 -> SETTING_SELECT 1 is missing, because LWL_INT/U16 was disconnected.

Reasons for the failure could be:

Disconnected or damaged FOC (LWL_INT/ U16 to RFIS U8), failure of component LWL_INT, a.s.o.

2.7.3 TAS_CM/ TAS_3TM_2 Functional Test

The test

- confirms the function of the TAS rf-path switches S1(Imaging/Service) and S2 (BC/LC).
- confirms the correct function of the 50 Ohm dummy load in the TAS_CM/ TAS_3TM.
- differentiates between malfunction of TAS or RFPA.

2.7.3.1 Required Tools

- Function Description RF system (FUN)
- Diagrams RF system
- Adapter BNC-NEG/N-POS (31 64 514), part of service tool kit
- Connector BNC-Banana plug (30 25 749), part of service tool kit
- 2m BNC cable (87 33 479), part of service tool kit
- Digital multimeter (DVM), type Fluke 187 or equivalent
- Service Plug kit, Avanto/ Espree (77 15 720).

2.7.3.2 Procedure

Step 1 - Test of TAS/S1 switch position and function of 50 Ohm dummy load

Prerequisites:

- The MR-System is ON and ready for measurement.
- The TX-path is switched to BC (no local coil is plugged in).

Checks:

- Check at TAS green LEDs 1 and 2 (power supply unit +/-15V) are ON.
- Check at TAS yellow LEDs 3 and 4 are OFF (TX-path switched to BC).

Measurement:

- Disconnect LC-cable at TALES /X6.
- Disconnect cable at BCCS/ X7, leave BCCS/ X7 open.
- Connect this cable at TALES /X6.
- Disconnect cable at TALES/ X5.
- Connect the adapter BNC-NEG/N-POS at TALES/ X5.
- Connect the 2m BNC-cable to the adapter and at the other end of the BNC-cable to BNC-Banana plug.
- Connect a DVM to the BNC-Banana plug and measure the resistance (range 200 Ohm).
 - » The measured resistance must be approximately 50 Ohm.
- **Note:**
After the measurement disconnect the adapter from TALES/ X5 and reconnect the original cable to TALES/ X5 again.

Conclusions

50 Ohm OK:

- » The TAS is correctly switched to Imaging/BC by S1 and the 50 Ohm TAS dummy load is OK.

50 Ohm not OK (open/ short circuit or significantly different from 50 Ohm):

- » If the green LEDs 1 and 2 are OFF, check the cable from TAS/ X7 to RFIS/ X10. Check the cable, the power supply unit, and the control signals from RFIS.

If OK:

The internal relay control board of the TAS is defective -> Replace defective TAS.

Tab. 24 Connector X7 at TAS_CTRL2 (examination room)

Pin	Name	Function
1	P15V	Input +15 V
2	GND	Input ground
3	N15V	Input -15 V
4	TAS_S1	+ Control input relay S1
5	TAS_S2	+ Control input relay S2

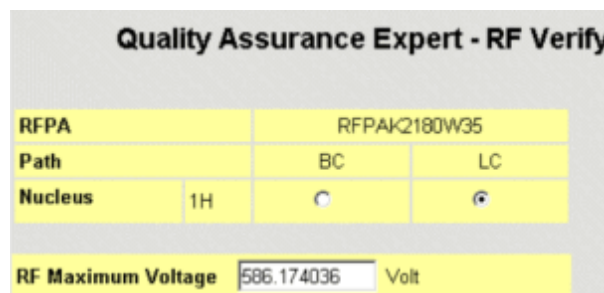
- » The TAS/ S1 has not switched to BC -> Replace defective TAS.
- » The 50 Ohm TAS dummy load is defective -> Replace defective TAS.

Step 2 - Test of TX-path LC (position of switch TAS/ S2) and function of RFPA

- The cable to BCCS/ X7 is disconnected at BCCS/ X7 and connected to TALES X6.
- Connect the Service Plug to coil plug #1 at patient table.
- Run **QA/ RF Verify LC** or if not available **QA/ Expert/ RF Verify** (on systems without spectroscopy option).

When using **QA/ Expert/ RF Verify** select the Nucleus 1H and LC-path in the Expert platform.

Fig. 23: QA/ RF Verify - Expertmode platform



- Check at TAS: LED 3 = OFF, LED 4 = ON. (If OK, the external control of TAS is OK).

Conclusions

RF Verify LC finishes successfully:

- » The TAS is correctly switched to Imaging/LC by S2 and the RFPA is OK. Proceed with step 3.

RF Verify LC fails:

- » The TAS has not switched to Imaging/LC by S2 or the RFPA is defective. Proceed with Step 3. If Step 3 confirms an intact RFPA, replace defective TAS.

Step 3 - Test of TX-path BC (position of switches TAS/ S1/ S2) and function of RFPA

- Disconnect RF-cable from TALES/ X6 and reconnect it to BCCS/ X7.
- Reconnect LC-Cable at TALES X6.
- Disconnect and remove the Service Plug and establish the measurement configuration for QA/ RFVerify: Position the large spherical phantom with loader in the magnet iso-center.
- Run **QA/ RF Verify**.
- Check at TAS:
 - LED 3 = OFF, LED 4 = OFF (first part of RF Verify).
 - LED 3 = ON, LED 4 = OFF (second part of RF Verify).
 - (If OK, the external control of TAS is OK).

Conclusions

RF Verify finishes successfully:

- » The TAS has correctly switched over to "service" by S1 during the execution of RF Verify and the RFPA is OK.

The first part of RF Verify (Calibration Voltage Search) fails:

- » The RFPA is probably defective. Check RF-output power using TestTools/RF/ RFPA. Replace defective RFPA.

The first part of RF Verify (calibration voltage search measured values from TALES) is ok but the second part fails (measured values from TAS dummy load):

- » The RFPA is OK. The TAS has not switched S1 to "service" (TAS dummy load). Replace defective TAS.

2.7.4 Excessive Reflected RF Power was detected by TALES

2.7.4.1 Error Message:

MRI_PAD_23 "MRI_PAD_PAD_PALI_TALESPPEAKR"

This message appears in a pop-up window for the customer and within SeSo -> EventLog. It is very likely caused by arcing components (e.g. bad cable connection/contact) in the transmit path (RF cables between RFPA and body coil).



To disable adjustments and/or safety supervisions (such as RFSWD or GSWD) during the following measurements, use "Tools" in the header bar of the user interface for Test-Tools. The service patient can also be de/registered in the "Tool-window."

Action 1:

1. Run QA/RF verify and TestTools AMC tests.
2. Run Tune-up/RF characteristic if QA fails.

Action 2:

Carefully check the cable connections between the RFPA and the body coil. Open all RF connectors between the TALES and body coil and inspect the inside of the connectors for black burn marks. Check if the RF connectors of the RFPA, ACC roof, and RF-filter plate are properly tightened.

RF-cable/connector checks:

- Are all connections tightened with the correct torque?

Tab. 25 Torque for RF-connections

Connector standard	Torque	Size open end wrench
N	1 - 2 Nm (1.7 Nm)	(manually, very tight)
HN	4 - 8 Nm (6 Nm)	SW 22
7/16	25 Nm	SW 27 and SW 32

- No metal shavings due to improperly attached cable connectors?
- Cables securely seated in plug?
- No black burn marks visible in RF connectors

If all ok and the problem remains, go to Action 3.

Action 3:

Run the tune-up procedures, "Tuning Calibration" and "BC-Tuning"; go to Action 4.

Action 4:

Try to reproduce the problem with sequence haste ("worst case")

1. To reproduce possible arcing, remove all local coils and position the small spherical phantom without loader. This results in a higher Q of the coil and therefore higher voltages.
2. Register the service patient and disable the RFSWD.
3. With the help of the Examination Manager, drag and drop the "haste" sequence from SIEMENS/sequence region/Siemens Sequences/Default protocols/haste. Open the sequence. Do not change any parameter in this sequence.

If the measurement aborts with the error message "Excessive reflected RF pulse power was detected by TALES", go to Action 5; otherwise go to Action 6.

Action 5:

1. Remove magnet side cover and rear side cover.
2. Set all three gradient DAC inputs to digital zero by setting the front switches S1/S2 on front of boards D10 (GSSU) to the off position (top). Remark: next measurement will be performed without pulsing gradients.
3. Disable GSWD.
4. In the contrast card, set measurements to 100.
5. Switch off the refrigerator/compressor and light in the examination room.
6. Start the measurement at the magnet panel.
7. Listen and look!

If arcing audible or visible at the BC or RF cables/connectors at the back of the body coil, replace the arcing component.

If the body coil is arcing, record the serial number before shipping back.

Switch gradient DACs back to their original settings.

Action 6:

Perform TestTools RF/BC tests functional and interactive. The specification for transmission: $T < -16$ dB. After proper adjustment of the Body Coil, it should be lower than -20 dB.

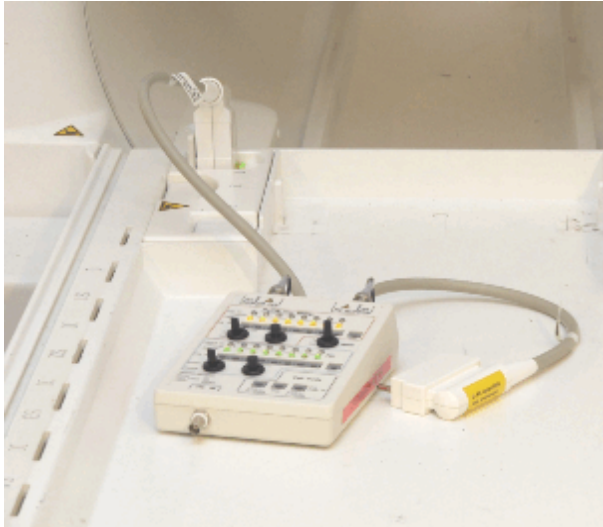
2.7.5 Check for broken LC-TX Cable

The LC-TX cable in the PTAB energy chain is exposed to mechanical stress during table movement. A broken LC-TX cable can cause the following symptoms:

- Ghosting and image artifacts with local TX-coils due to instable transmit power
- Measurement aborts with local TX-coils due to inconsistent/ unstable transmit power
- Excessive reflection detected in the LC-TX path

Use the following procedure to check for a broken LC-TX cable:

Fig. 24: Service Plug connected



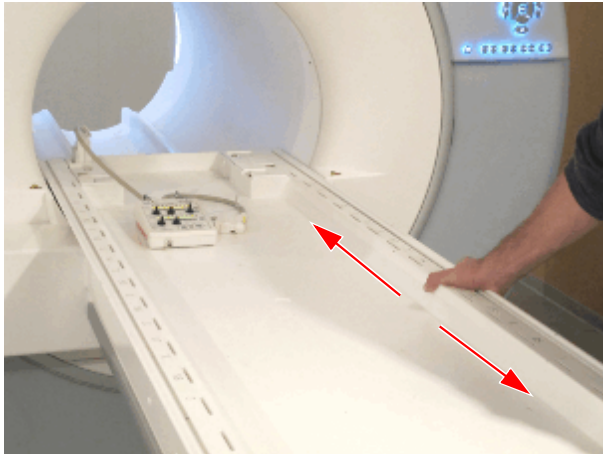
- Connect the Service Plug to coil plug 1

Note:

Make sure, the BNC connectors at the Service Plug rear-side are properly connected (internal 50-ohm terminator).

- Run **SeSo > TestTools > RF > Cable LC**.
- Evaluate Test Report

Fig. 25: Move table back and forth



In case the above single test shows no evidence, run repeated tests while moving the patient table by hand:

- Press Table Stop and/ or release the horizontal emergency clutch of the table
- Run **SeSo > TestTools > RF > Cable LC** and set the **Repeat Count** for multiple measurements, e.g. 4.
- During the test, move the table back and forth by hand, checking the whole movement range.

Fig. 26: DVM adapters

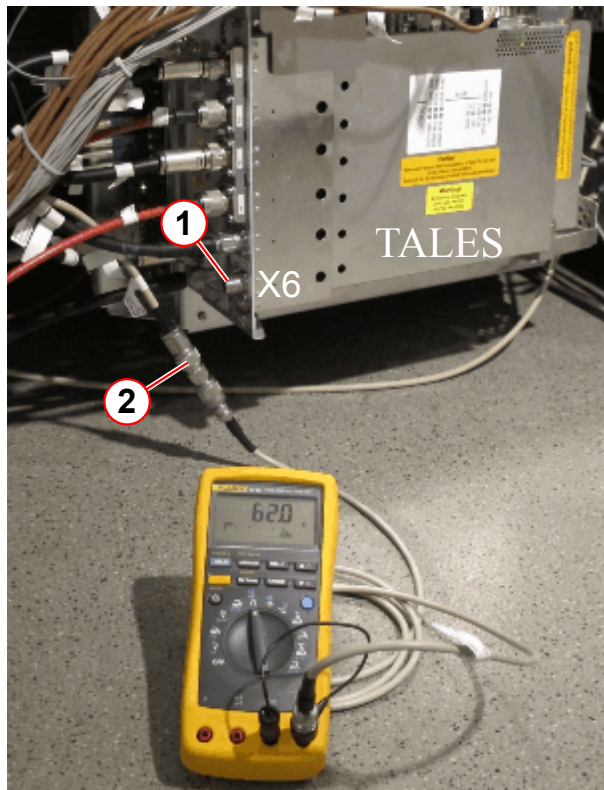


- (1) Connector BNC-NEG/2BAN (3025749)
- (2) RF-Cable BNC 2 m (8733479)
- (3) Adapter 1P N-BNC (03164514)
- (4) Connector N-BU/N-BU

Optional test method with DVM:

- Configure the DVM adapters

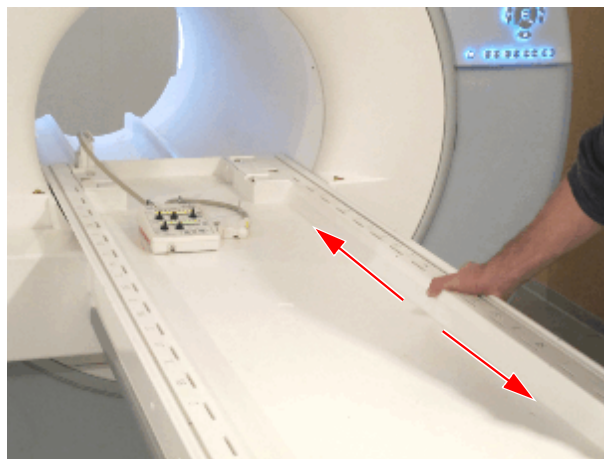
Fig. 27: DVM connected to LC-TX cable



- (1) Disconnect LC-TX cable from TALEX X6
- (2) Connect DVM to LC-TX cable end

- In the magnet room, disconnect LC-TX cable from TALEX X6.
- Connect the DVM to the LC-TX cable end.

Fig. 28: Move table back and forth



- Press Table Stop and/ or release the horizontal emergency clutch of the table.
- Move the table back and forth by hand, checking the whole movement range.

Fig. 29: DVM check 50 ohm



- During table movement, read the value at the DVM
 - » A stable value of approx. 50 ohm should be displayed.
 - » If the value is unstable, or a short or open circuit is measured, the LC-TX cable is broken.

2.7.6 Check of 50 Ohm/Short Service Adapters (failed Tuning Calibration)

NOTICE

Use of defective 50 Ohm/Short service adapters leads to invalid Tuning Calibration results.

Failing or invalid Tune-up step Tuning Calibration or BC Tuning. Reduced TX-performance of the system.

- » For Tim-systems, the Tune-up step Tuning Calibration will fail completely when defective RF-service adapters are used.
- » Nevertheless, with minor deficiencies, Tuning Calibration can pass but provides wrong calibration data or leads to a failing BC Tuning.
- » Before use, always check condition and measure resistance of all service adapters/cables with a DVM.
- » In case of any deficiencies, order/use new service adapters.

2.7.6.1 Problem Description

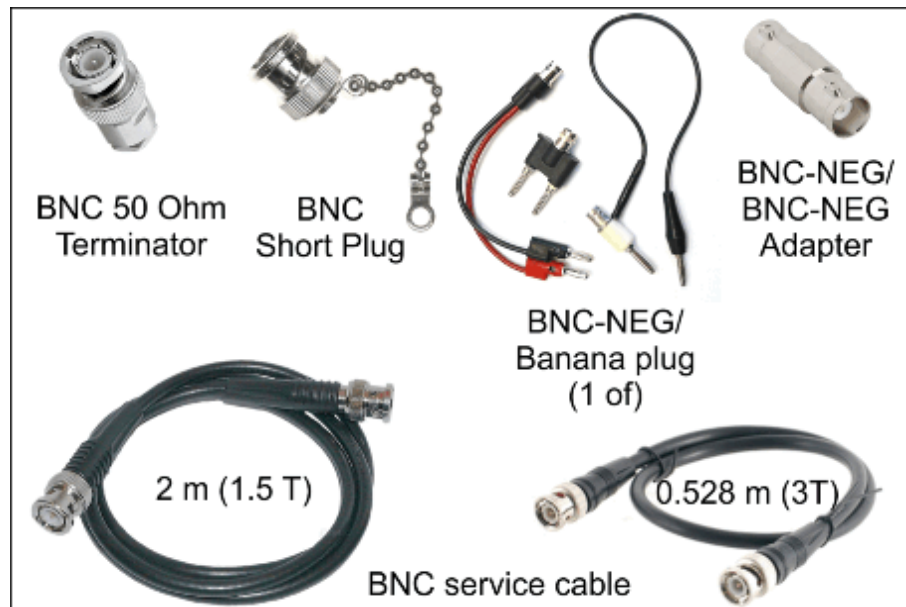
Use of defective RF service adapters can lead to multiple problems which are difficult to troubleshoot, e.g.:

- Failing Tune-up step Tuning Calibration (due to e.g. wrong resistance value of 50 Ohm terminator, broken BNC service cable, etc.).
- Unnecessary spare part consumption (e.g. new TALES sent back as "dead on arrival" due to failing Tuning Calibration).

- Wrong display of BC Tuning values due to bad Tuning Calibration (Tuning Calibration is the reference for BC Tuning).
- Reduced TX-performance of the system due to wrong BC Tuning Adjustment as a result of a bad Tuning Calibration.

2.7.6.2 Required Tools

Fig. 30: Required Tools



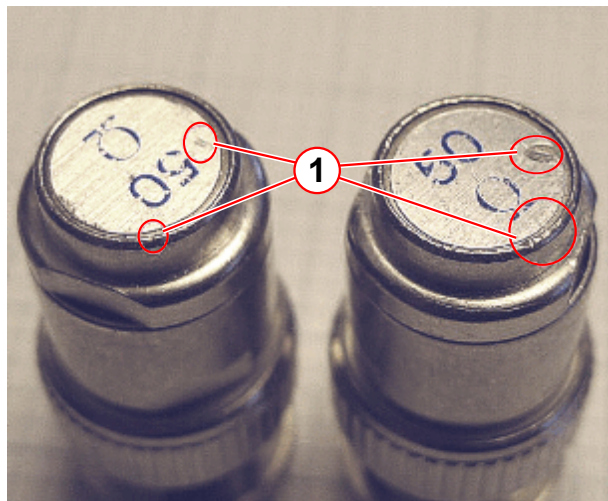
- Digital Multimeter FLUKE 287 or equivalent with a valid calibration
- Adapter BNC-NEG/Banana plug (30 25 749), part of service tool kit
- Adapter BNC-NEG/BNC-NEG (77 60 481), part of service tool kit
- 2 x BNC 50 Ohm terminator (31 10 392)
- 2 x BNC Short plug (31 63 714)
- BNC service cable used for Tune-up step Tuning Calibration:
 - BNC 2 m cable for 1.5T systems (87 33 479)
 - BNC 0.528 m cable for 3T systems (109 15 732)

2.7.6.3 Visual Checks

- Check condition of cables/adapters visually:
 - » There must be NO damage or excessive wear.
 - » Only use adapters which are in a good condition.
 - » Below, there are some examples for connectors which are NOT ok:

Examples of defective adapters → replace:

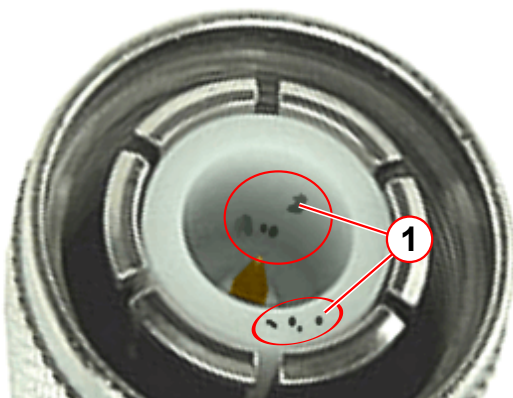
Fig. 31: 50 Ohm BNC Terminators



(1) Notches

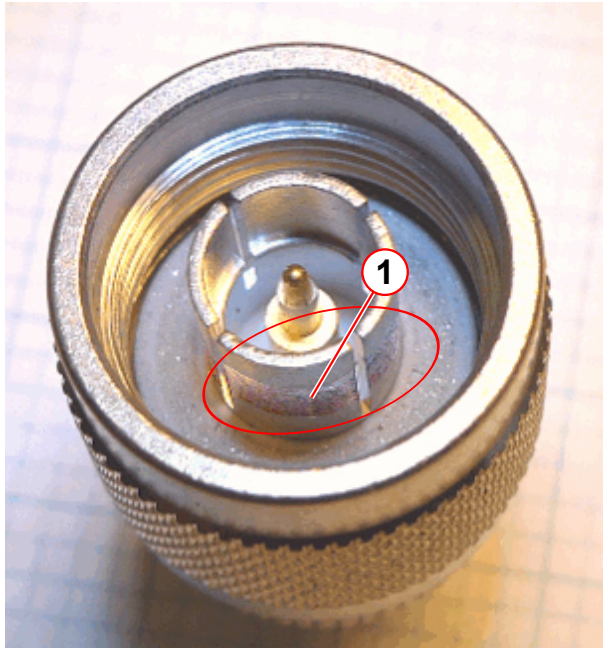
- Signs of mechanical damage, e.g. notches at dropped adapters:
 - » Handle adapters carefully.
 - » Store and transport adapters in a suitable box (e.g. with foam protection).

Fig. 32: BNC-HN Adapter

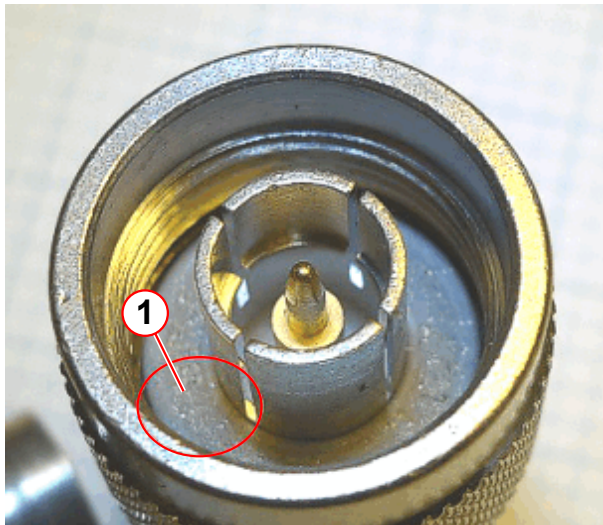


(1) Dirt

- Contamination with dirt, liquids, etc.:
 - » Work in a clean environment.
 - » Store, transport adapters in a protective packing.

Examples of defective adapters -> replace:*Fig. 33: BNC Adapter*

(1) Scratches outer conductor

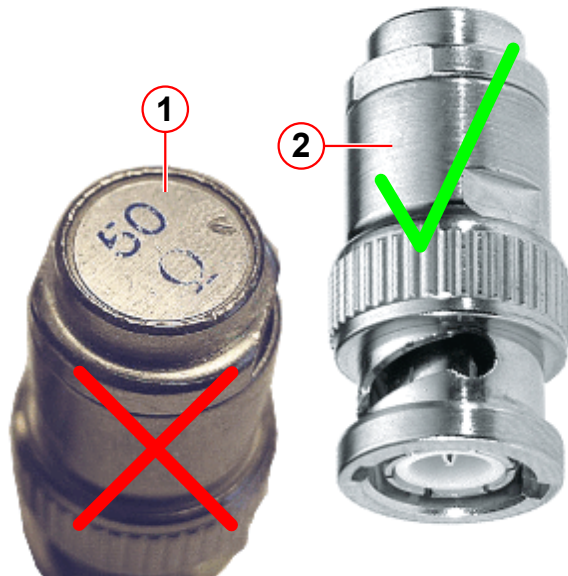
Fig. 34: BNC Adapter

(1) Metal dust

- Connector metal abrasion, e.g. scratches at outer conductor:
 - » Hold adapters when tightening - avoid turning.

Examples of defective adapters -> replace:

Fig. 35: 50 Ohm BNC Terminator



- (1) Oxidized - do NOT use!
(2) Shiny - good condition

- Oxidization:
 - » Scrap old, oxidized adapters - only use new ones.

2.7.6.4 Measurement Procedure with DVM

Step 1:

Fig. 36: Resistance measurement 50 Ohm BNC terminator



Measure resistance of **BNC 50 Ohm** terminators:

- Use DVM with adapter BNC-NEG/Banana plug.
 - » The resistance must be 50 ± 0.5 Ohm.
 - » Otherwise, replace adapter before use.

Step 2:

Fig. 37: Resistance measurement BNC Short Plug



Measure resistance of **BNC Short Plug**:

- Use DVM with adapter BNC-NEG/Banana plug.
 - » The resistance must be $0 \pm 0.5 / - 0$ Ohm.
 - » Otherwise, replace Short Plug before use.

Step 3:

Fig. 38: Resistance measurement BNC Service cable

Measure resistance of **BNC Service Cable**:

- Use DVM with adapter BNC-NEG/Banana plug.
- Connect service cable to DVM adapter.
- Connect adapter BNC-NEG/BNC-NEG to the cable end.
- Connect BNC Short Plug.
 - » The resistance must be $0 + 0.5 - 0 \text{ Ohm}$.
 - » Otherwise, replace BNC Service Cable before use.

2.7.7 RFPA Power Fault / RFPA Overcurrent

2.7.7.1 Error Messages:

Avanto Systems:

MRI_SLO_27MRI_SLO_CANSLIO_RFPAPOWERFAIL

MRI_SLO_30 MRI_SLO_CANSLIO_RFPACURRENT

MRI_SLO_31 MRI_SLO_CANSLIO_RFPARFPOWER

Avanto Dot Systems:

MRI_RFP_2002

MRI_RFP_2005

Action 1:

Check the ownership of the sequence:

1. Drag and drop the critical Customer Protocol with the help of the Examination Explorer and open it.

2. Manually set the RF amplitudes: click the Parameter card on System and set each RF amplitude to zero in the displayed Transmitter/Receiver card.
3. Start the measurement. After the measurement is finished, call up image(s) and check the image text.

If there is a *sign visible in front of the sequence file name, go to Action 2. Otherwise, this is not a Siemens-released sequence, which may require more RF power as specified for the RFPA.

Action 2:

Measure supply voltage of RFPA (L1-N, L2-N, L3-N, 93 +/- 5.6V) only with a DVM (Fluke).

Attention: Do not measure the voltage with a scope!

If the voltage is OK and stable, but the problem occurs again during patient imaging, go to Action 3. Otherwise, adapt the line transformer correctly and provide for stable line supply.

Action 3:

1. Position the large spherical phantom with loader.
2. Select official Customer Protocol.
3. Open a Command Prompt window. Get access to "percom" (to apply RF to the dummy load in the TAS): After typing in "percom" in the command prompt window, enter a password. Call your local USC or HSC for generating a password. In the menu displayed, choose **RFISK2254 (1)**. In the menu displayed, choose **Set TASPath (25)**.
4. Right-click the worker icon to open a pull-down menu. Select **Properties** in the displayed window, then **"Protocol Properties"**. Click **"Execute Properties"** or **"Execution"** and activate the **"Wait user to start"** checkbox. Click "OK". Open the sequence (no measurement!).
5. Start preparing for measurement with "Apply". Do not click **"Continue"** in the displayed **"Load/Start Protocol"** window!
6. At the command prompt (percom), enter: Enter the path description for switch S1 ([unchanged], dummy load, imaging): **dummyload <Return>**, Enter path description for switch S2, coil ([unchanged], body, local) **body <Return>**
7. Check at TAS that LED S1 is on and LED S2 is off.
8. Start the measurement, clicking **"Continue"** in the window **"Load/Start Protocol"**.
Note: After each sequence runs, the default TX-path (Imaging/Bodycoil) is active.

If the measurement aborts, go to Action 4, otherwise to Action 5.

Action 4:

Disconnect the RF transmit cable at the RFPA output and measure the resistance of the cable (center line versus cable shield). If higher than 50 ohms, check the cables/connections to the TAS; check TAS_3TM. Run TestTools/RF/Cable using RF adapter set 73 85 896.

Action 5:

Perform TestTools/RF/BC (functional/interactive) tests and contact CS HSC Erlangen or USC Cary.

2.7.8 RFPA Overtemperature

2.7.8.1 Error Message:

Avanto Systems;

MRI_SLO_28 "MRI_SLO_CANSLIO_RFPATEMPERATURE"

Avanto Dot Systems:

MRI_RFP_2003

Action 1:

Check the ownership of the sequence:

1. Drag and drop the critical customer protocol with the help of the Examination Explorer and open it.
2. Manually set the RF amplitudes: Click the System parameter card and set each RF amplitude to zero on the displayed transmitter/receiver card.
3. Start the measurement. After the measurement is finished, call up image(s) and check the image text.

If there is a *sign visible in front of the sequence file name, go to Action 2. Otherwise, this is not a Siemens-released sequence, which may require more RF power as specified for the RFPA.

Action 2:

Start the sequence. If the measurement aborts during protocol loading or 1 to 2 seconds after start, check water flow and temperature.

If ok, replace the RFPA in consultation with CS HSC or USC Cary. Otherwise, go to Action 3.

Action 3:

Check water flow through the RFPA. Check for arcing components, see the procedure, "Excessive reflected pulse power". Perform TestTools/RF/BC (functional/interactive) tests.

If ok, replace the RFPA in consultation with CS HSC or USC Cary.

2.7.8.2 Checking the RFSWDHistoryList*.log

This file can be accessed with SeSo/Reports/Startup Process/"Process diagnostics".

Whenever the patient is changed (Pat/Reg) and the first protocol is started, a new file RFSWDHistoryListNew.log is generated and the original file is copied to RFSWDHistoryListOld.log.

In the case of high RF reflection, for example, the file is also saved as RFSWDHistoryListErr.log. Select one of these files and scroll down to line number 1000 (Consistency Check Data - after measurement). In line 1011, the peak maximum forward and peak maximum reflected power measured by TALEs channel 1 are shown in watts. In line 1012, the same values measured by TALEs channel 2 are shown.

Example (ok condition!):

1011

PALIPeakPowerForw 0.....: 1424.420654 [W]

PALIPeakPowerRefl 0..... : 34.159729 [W]

1012

PALIPeakPowerForw_90.....: 1374.220337 [W]

PALIPeakPowerRefl_90..... : 9.716617 [W]



The absolute power values depend on the selected sequence. The forward/reflected ratio of each channel indicates the quality of impedance matching.

2.7.9 Checking Water Flow through RFPA

Open the water connection at the RFPA (DORA) output and measure how many liters flow into a bucket in 1/2 min. Given the nominal value of 8 l/min, it should be at least 4 liters.



Two quick-release water couplings are needed to conduct this test: Material number 5940726 (male coupling) and material number 5940718 (female coupling).

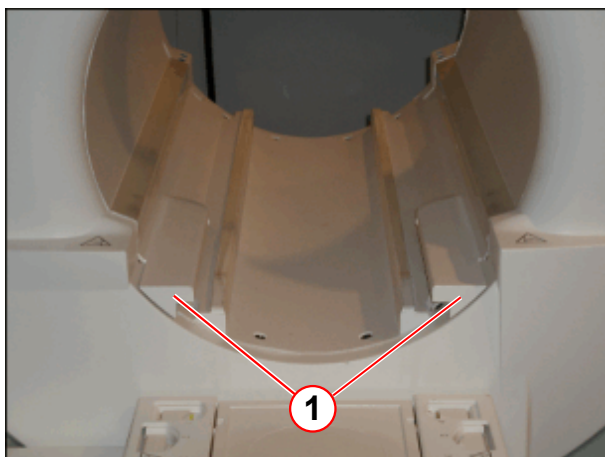
Refill the cooling circuit, if necessary. A hand pump (service tool) or water pump with drill is required, when refilling the drained 38% ethylene glycol water mixture. (→ Drain procedure of the ACS / M6-020.840)

2.7.10 Repair of Table Guide Flange inside the Body Coil

The table guide flange is glued and bolted to the body coil.

If the front end of the table guide flange was torn off, it can be glued back to the body coil again.

Fig. 39: Table guide flange



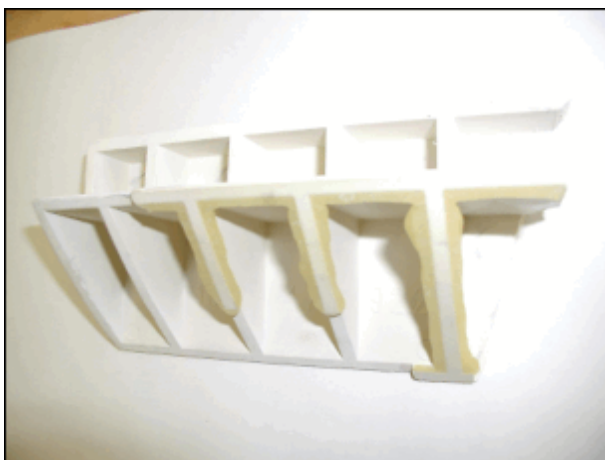
(1) Table guide flange

Use an instant adhesive for plastic.

The adhesive surface must be clean, greaseless, and free of dust.

Depending on the adhesive used, the torn-off piece should not be mechanically stressed for some time. To minimize the visible remaining clearance between the glued piece and the rest of the table guide flange, use the set of touch-up paint delivered with the system (material number 8117934).

Fig. 40: Torn off piece



2.8 RF Test Procedures using Test Tools

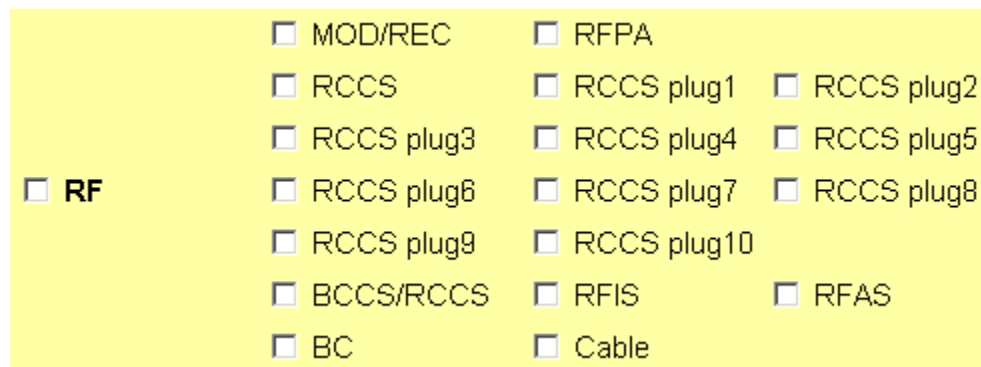
2.8.1 General

RF test procedures using TestTools in Software VBxx:

This section describes additional tests for troubleshooting components (FRUs) of the RF system, such as components in the TX-path (RFPA, TALES, BC and RF cables).

- Tools (switch off supervisions, additional options)
- RF/BC
- RF/Cable

Fig. 41: TestTools RF Tests



2.8.1.1 RF Test Procedures using TestTools in Software Version VDxx

Category	Function	Test Name	Description (refer also to Online Help)
Fast	TX	Max Power	Check of maximum RFPA power output
Fast	TX	Power Dico TALES	Check of maximum rf-power send to BC, power reflection and deviation of DICO and TALES values
Fast	TX	BCCS Transmit/ Receive switch	Check of the Transmit/ Receive Switch in the BCCS
Fast	TX	TALES Reference Voltage	Check of the reference voltage of the TALES
Fast	RX	Receiver	Check of receive signal amplitudes of all RX channels
Normal	TX	TX Stab/ Lin	Check of the linearity and short time stability of the RFPA with transmission path set to dummy
Normal	TX	BC Tune/ Detune	Check of the BC PIN Doides

Category	Function	Test Name	Description (refer also to Online Help)
Normal	TX	BC Tuning	Reflection and transmission of the two BC parts as a function of the frequency (like QA function)
Normal	RX	RX Stab/ Lin	Check of the linearity and short time stability of the receive paths
Normal	RX	RCCS Matrix	Check of all possible signal paths (nodes) in the RCCS
Normal	RX	RCCS SGA	Check of the linearity and short time stability of the RCCS SGAs
Normal	RX	BCCS Stab/ Lin (0 degree forward)	Check of the linearity and short time stability of the BCCS SGAs and the BCCS receive hybrid
Normal	RX	BCCS Stab/ Lin (90 degree forward)	Check of the linearity and short time stability of the BCCS SGAs and the BCCS receive hybrid
Interactive	TX LC	Power LC interactive	Check of the RF power transmit path for a local 1H TX coil.
Interactive	Cable BC	Cable BC-TX-Path	Checks for defective cables in the BC TX chain. A dummy load can be connected at different points of the BC TX chain. A pop-up is displayed that requests the point where the dummy is connected.

Fig. 42: TestTools/ Expert/ Fast

☐ TX	☐ Max Power
	☐ Power Dico Tales
	☐ BCCS Transmit Receive Switch
	☐ TALES Reference Voltage
☐ RX	☐ Receiver

Fig. 43: TestTools/ Expert/ Normal

☐ TX	☐ TX Stab/Lin
	☐ BC Tune/Detune
	☐ BC Tuning
☐ RX	☐ RX Stab/Lin
	☐ RCCS Matrix
	☐ RCCS SGA
	☐ BCCS Stab/Lin (0 degree forward)
	☐ BCCS Stab/Lin (90 degree forward)

Fig. 44: TestTools/ Expert/ Interactive

☐ TX LC	☐ Power LC interactive
☐ LC Plug 1	☐ LC Plug RX Plug 1
	☐ LC Plug Pin Plug 1
☐ LC Plug 2	☐ LC Plug RX Plug 2
	☐ LC Plug Pin Plug 2
☐ LC Plug 3	☐ LC Plug RX Plug 3
	☐ LC Plug Pin Plug 3
☐ LC Plug 4	☐ LC Plug RX Plug 4
	☐ LC Plug Pin Plug 4
☐ LC Plug 5	☐ LC Plug RX Plug 5
	☐ LC Plug Pin Plug 5
☐ LC Plug 6	☐ LC Plug RX Plug 6
	☐ LC Plug Pin Plug 6
☐ LC Plug 7	☐ LC Plug RX Plug 7
	☐ LC Plug Pin Plug 7
☐ LC Plug 8	☐ LC Plug RX Plug 8
	☐ LC Plug Pin Plug 8
☐ LC Plug 9	☐ LC Plug RX Plug 9
	☐ LC Plug Pin Plug 9
☐ LC Plug 10	☐ LC Plug RX Plug 10
	☐ LC Plug Pin Plug 10
☐ Cable BC	☐ Cable BC-TX-Path

2.8.2 Tools

2.8.2.1 TestTools/Tools

The Tools button enables you to access some special settings of the measurement system. Previously, this was only possible via separate passwords and the operating system level:



It is possible that the message "Invalid value for Adjustments mode: Othermode" is displayed when entering the platform for the first time during a test session.

This "error message" is correct since other modes exist besides the listed modes for the Service Software. However, these modes cannot be displayed or selected. These other modes are used, e.g., in Tune-Up and QA.

Please set the new mode according to your needs.

The following radio buttons are available:

- The watchdog programs for RF (RFSWD), gradient (GSWD), and the adjustments can be set to
 - Patient mode (called "enabled" in former software versions)
 - Service mode (called "disabled" in former software versions)
 - "Disabled" (only RFSWD, to this extent, not available in former software versions)

NOTICE

When you set the RFSWD to "disabled", any protection for the components is completely switched off. This is normally not necessary.

Incorrect use of this setting may damage the system or parts of it.

- » **"Disabled" should not be used during normal troubleshooting! It should be used only if advised by the MR-Hotline (HSC).**

The setting of the RFSWD (Patient mode/Service mode) has no influence in the Test-Tools measurements as the tests already check the necessary prerequisites and set the mode accordingly.

- RF characteristic can be toggled between the measured characteristic and the linear one. However, this will affect only a few tests and is mainly intended for expert use.
- It is possible to register or unregister the Service Patient so that it is not necessary to run a test or QA to have this patient registered for further troubleshooting or service.
- **Avanto and Espree Systems:**
When a 10 V preamplifier error occurs, the behavior of the system can be changed from "one time reset" to "no reset" for troubleshooting (Autorecovery RCCS 10V Error). For customer use, the 10 V error is normally automatically reset one time to avoid system reboots. It may be necessary to disable this functionality for troubleshooting.

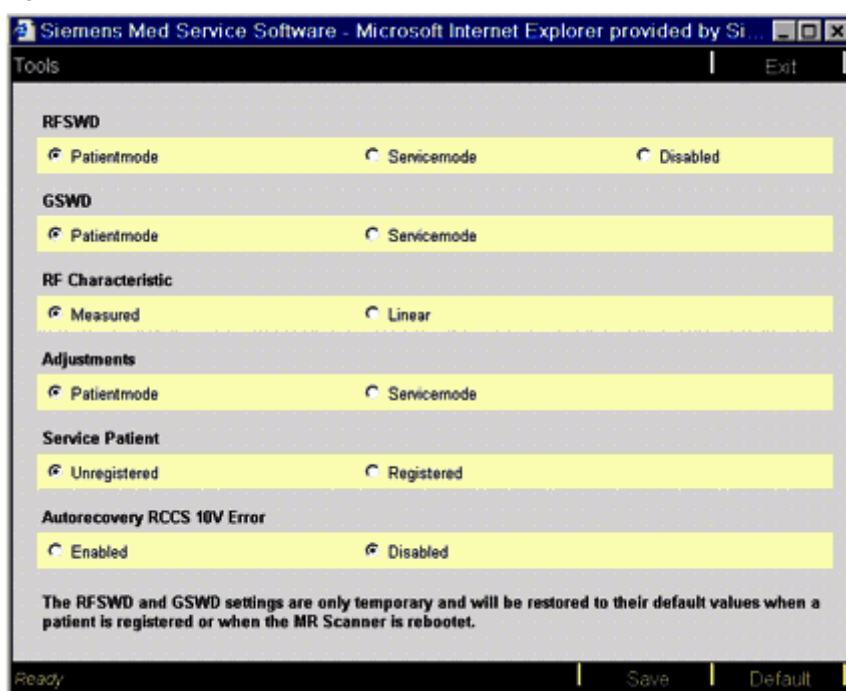


The setting of this functionality is reset to patient mode with each system reboot.

It may be helpful to change these settings during troubleshooting.

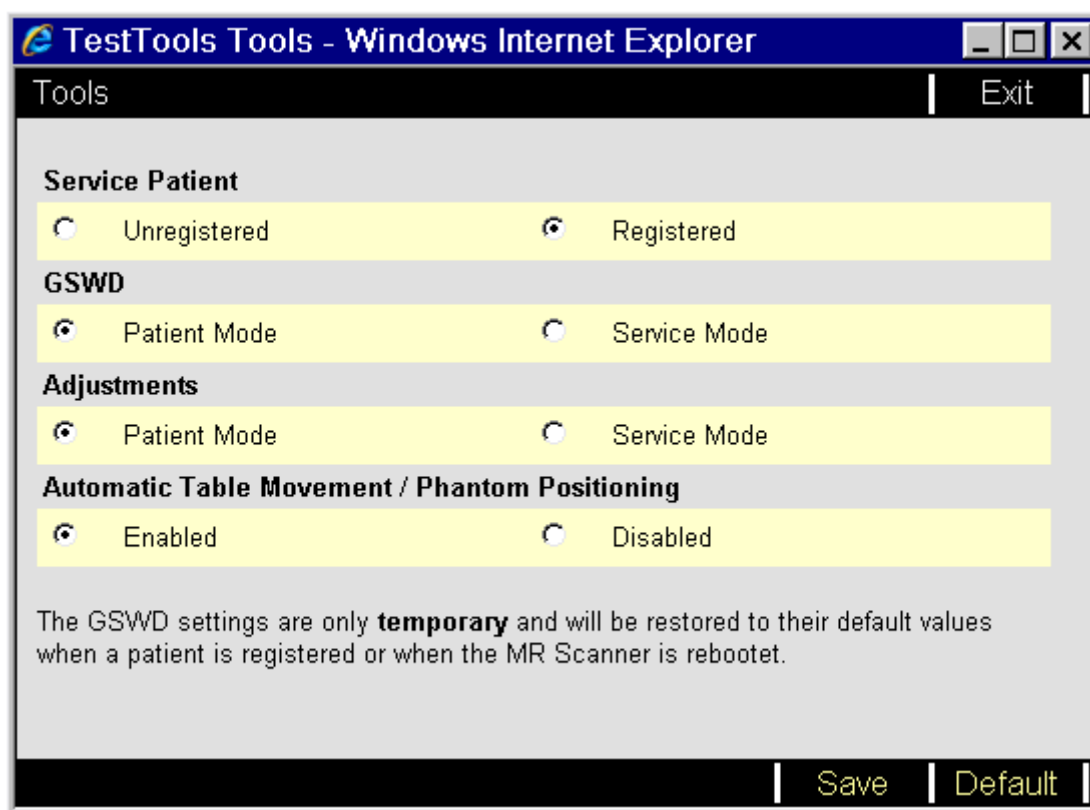
In the bottom action bar, the buttons "Save" (save the settings) and "Default" (reset to normal) are available, in the top it is possible to close the window via "Exit."

Fig. 45: TestTools/ Tools user interface window



Since Software Version VDxx the TestTools "Tools" window has changed:

Fig. 46: Example TestTools "Tools" Window



2.8.3 BC Tests

(For Avanto Dot Systems use TestTools/ Normal/ TX/ Bc Tuning)

2.8.3.1 Prerequisites

- Valid tuning calibration; TALES calibration is sufficient
- TestTools test must be ok: MOD/REC, RFAS loops
- The BC tests (except the BC power test) use low-voltage RF signals using the TTX1 output of the TX module via RCCS, BCCS back to the RX module.

2.8.3.2 Test Procedure

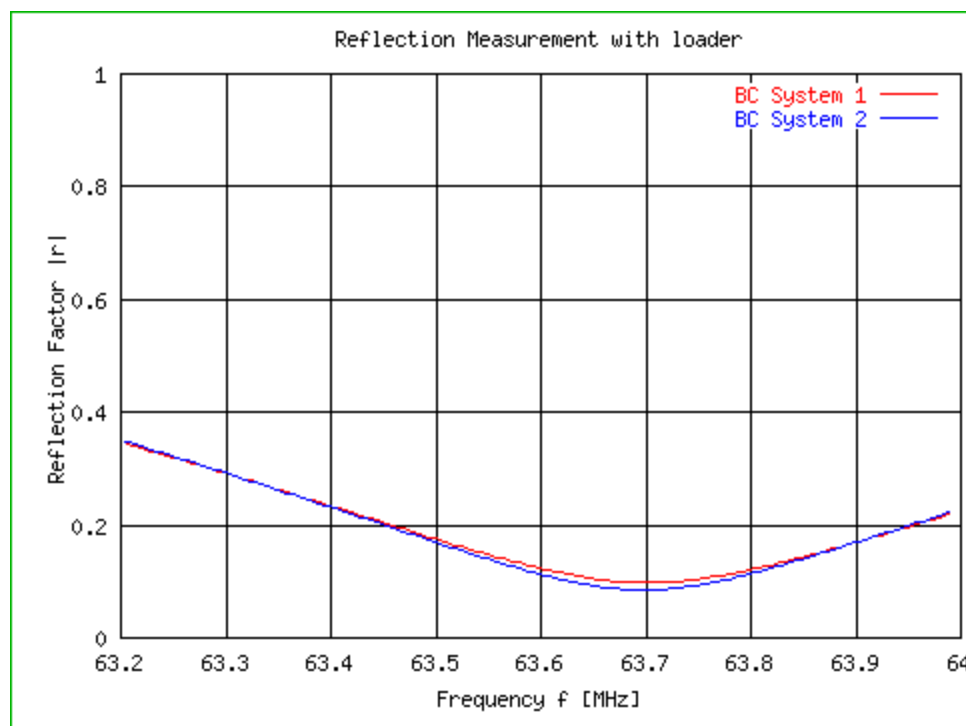
The basis for the tests is the measurement of reflection factors which are complex numbers.



For functional tests, ensure that the loader with the large spherical phantom is precisely positioned in the body coil.

- **Body coil functional (with phantom and load)**

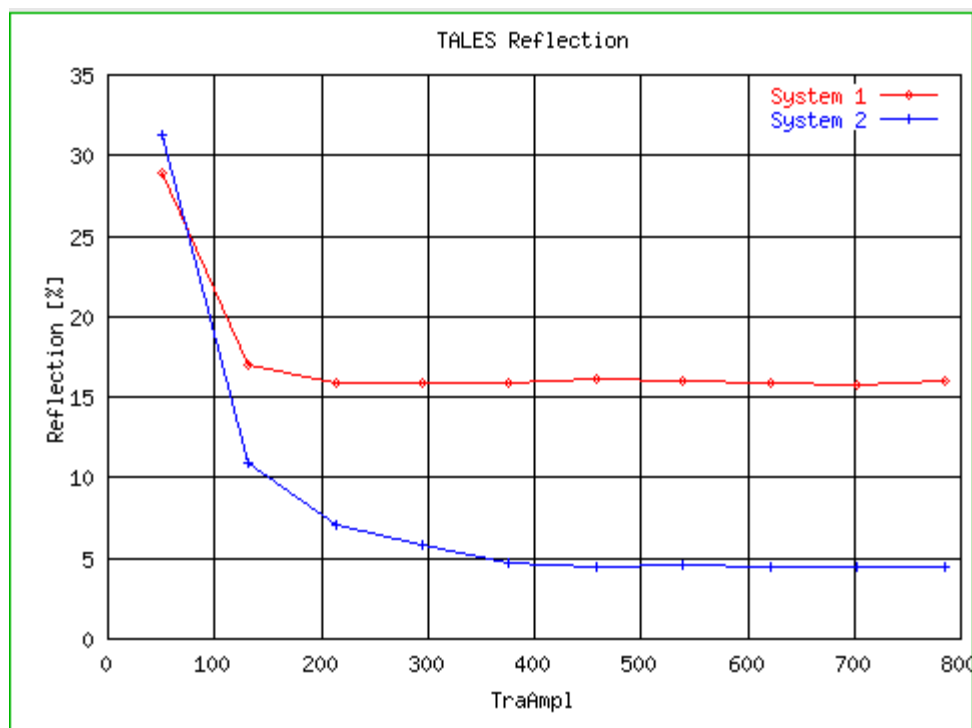
Fig. 47: BC - Reflection Evaluation (with load)



The body coil test is based on the analysis of reflective voltage factors that depend on the frequency. The test simulates a network analyzer that provides the frequency for the smallest reflection factor and the resonance curve of the body coil as results. The available frequency range of approximately 1MHz is stepped through in 10kHz steps. The respective reflection factors are produced and entered in a frequency reflection factor diagram. The frequency with the smallest reflection factor is checked against specifications. The Q-factor of the coil will be evaluated and checked against specification.

Additionally, a power test with increasing transmitter amplitude is performed. The TALEs reflection values over the transmitter amplitudes are graphically displayed.

Fig. 48: BC power test - TALEs reflection (with load)



■ Body coil interactive (without phantom and load)

The first part of the measurement is identical to the functional measurement.

Additionally, the decoupling of the body coil elements (transmission measurement) is evaluated.



For body coil adjustments, refer also to the manual, "Replacement of Parts", "RF-System", replacement of the body coil and tune-up procedure (→ BC Tuning / M6-020.841.14).

Fig. 49: BC - Reflection Evaluation (without load)

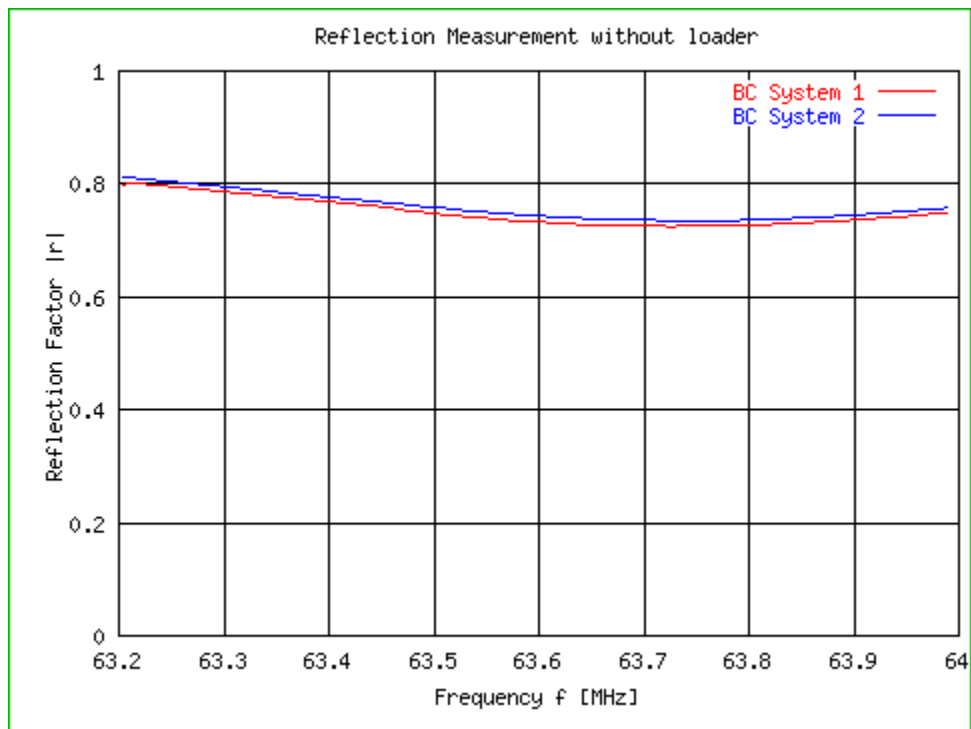
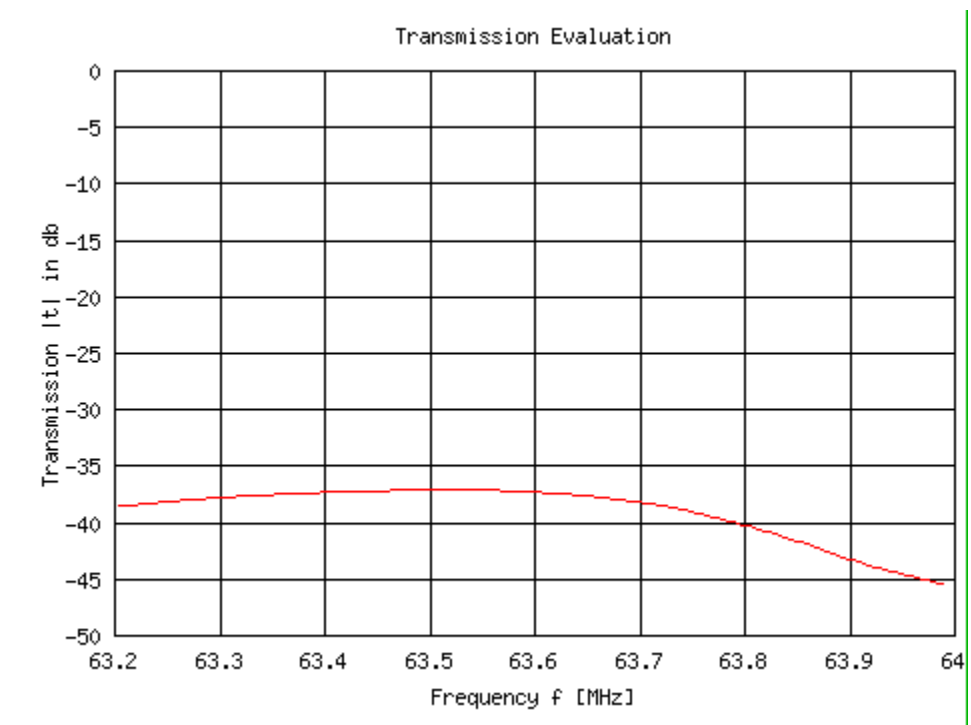
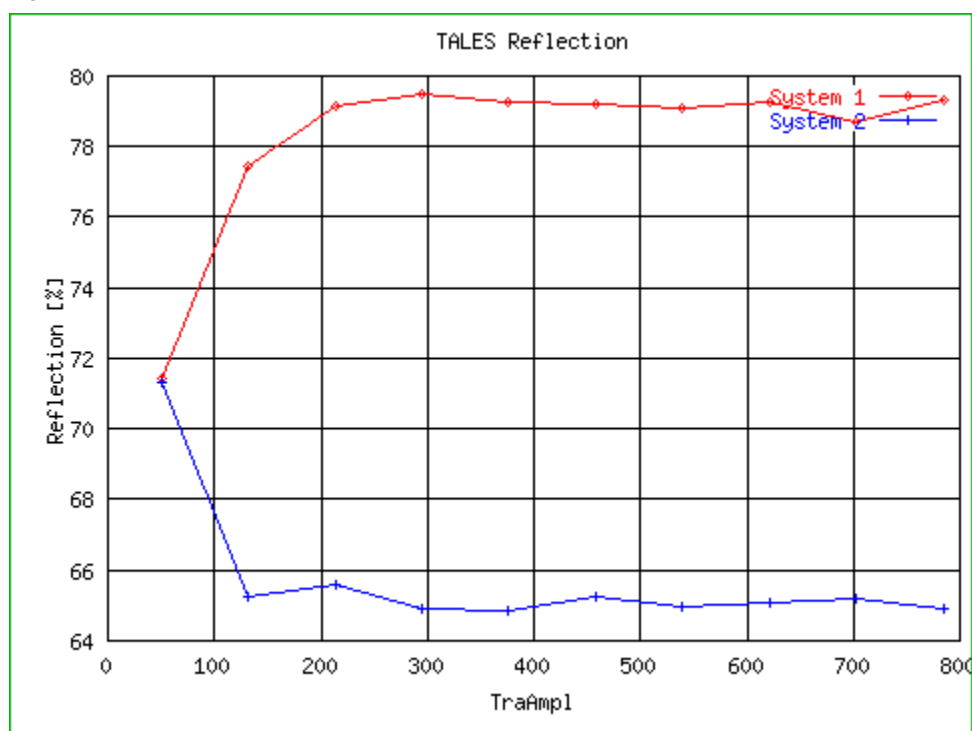


Fig. 50: BC - Transmission Evaluation (without load)



Like the functional test, a power test with increasing transmitter amplitude is performed. The TALES reflection values over the transmitter amplitudes are graphically displayed.

Fig. 51: TALEs reflection without load



Conclusions:

If the tolerances of the reflection measurement evaluation are exceeded, the problem is probably located in the BC.

The power test checks the disruptive strength of the BC. The test should run without shut-down to the maximum voltage. The reflection of TALEs should be approximately constant at voltages higher than 200 V.

The power test without load is the worst case, i.e. the reflected voltages are significantly higher.

2.8.4 Cable

- BC-TX path (refl.) (Avanto)
- BC-TX path (volt.) (Avanto)
- LC-TX-path (also refer to the on-line help) (Avanto)
or TestTools/ Interactive/ TX LC and Cable BC (Avanto Dot)



These tests are for experts only! Additionally, special equipment is necessary for these tests. Special tools (power dummies, RF cables, connectors, adapters and attenuators) are needed.



Whenever a dummy load is used to interrupt the TX-path BEFORE the TALES, the comparison of DICO- and TALES-values will not be successful and the test will fail.

This is normal and not a problem with the system

The tests enable you to check the complete TX-path to BC or LC and to systematically troubleshoot the TX-path by connecting a 50-Ohm terminator at various locations in the TX-path and/or configuration of particular loops.

The test should be used by experienced service engineers only. Special tools (power dummies, RF cables, connectors, adapters and attenuators) are needed.

Pop-up windows inform you about the actual measurement configuration.

2.8.4.1 Additionally required Tools:

- RF Adapter set (73 85 896)

2.8.4.2 Safety



WARNING

High RF voltage!

Contact with RF conductors during transmitting of the RFPA will result in death or serious injury.

- » Before disconnecting RF transmit cables or connectors in the transmit path, disconnect the RF input and the unblank line U1 from the RFPA.

NOTICE

Be careful not to overload the signal inputs when establishing non-standard measurement loops, e.g. when service sequences such as rf_loop and/ or rf_pulse will be used.

Choose the transmitter voltage, repetition time and pulse time so that the dummy load and attenuators are not overloaded.

- » Always start with low transmitter amplitudes (approx. 50V). The pulse amplitudes must be checked/ set for each new measurement.

2.8.4.3 Conclusion BC-TX-path (refl.)

If the reflections in the BC-TX path are too high, you can localize the defective component in the TX-path. Repeat the test after connecting the dummy load to a component in front of its current position. Check for loose connectors, in particular.

2.8.4.4 Conclusion BC-TX-path (volt.)

If the deviation between DICO and TALES is out of specification, repeat the RF characteristic measurement.

Locate the defective component in the TX-path by repeating the test after connecting the dummy load to a component in front of its current position.

2.8.4.5 Examples of Breakpoints in the TX- Path

Tab. 26 Breakpoints in TX-path

Location of dummy termination in the TX path	Symptom	Possible error	Service action
RFPA out (X2)	Power output too low	RFPA, DICO	Replace RFPA
ACC Top out (X 38)	High reflection DICO	RF cable ACC	Replace RF cable
TAS_3TM out (X4) TAS_3TM out (X5)	No signal/ low power	TAS_3TM control TAS_3TM switch	Check TAS_3TM power, TAS_3TM control (LEDs). For path switching, use percom. Check resistance of switch path with an ohmmeter. Replace defective TAS_3TM
BCCS in (X1)	High reflection DICO	RF cable/ connectors	Replace RF cable/ connectors
BCCS out (X2, X8)	Low power High reflection DICO	Defective T/R switch	Replace BCCS
TALES out (X3, X4) TALES out (X6)	High reflection TALES DICO OK	Oscillation of RFPA/ RFPA transmits not only MR frequency	Check RFPA, TAS_3TM, BCCS, cables, TALES
	High reflection TALES and DICO	Cable connections TAS_3TM to BCCS to TALES TALES and/or TALES control defective	Check cable connections. Check TALES control (reference voltage, FOC control, CV signal). Replace defective TALES.
	OK with dummy termination	BC error	Use TestTools, BC tests.

2.8.4.6 Troubleshooting Hints

Transmit Power too low

- Check TX module

- Check RFPA

Reflection too high

- Arcing of capacitors in BC
- Defective RF cables at BC and/or loose, corroded connections

Arcing

- Sporadic arcing -> BC defective
- Jumps in the reflection factor r with increasing transmit voltage
- $r > 70\%$ -> arcing in the BC

Imaging	Reflection (voltage) < 50% (with BC)
Termination 50 Ohm	Reflection (power) < 10% (with 50 Ohm dummy)

Note:

- Arcing is permanent: Arcing of C's in BC

Oscillation of RFPA

- The reflection measured by TALES in a transmit voltage interval is significantly higher than the reflection factor measured by DICO.

Reason:

TALES measures wide-band, DICO measures narrow-band

- » Oscillation of RFPA; the RFPA also transmits at other frequencies besides the MR frequency.

2.8.5 Using the 50 Ohm Termination of the Service Plug (LC-TX-path)

Using TestTools while start-up, maintenance, and troubleshooting for testing the local coil transmit path (LC-TX-path) with a Service Plug will result in failing tests in case the internal 50 Ohm termination resistor is defective.

For e.g. Tim systems (Avanto/Espre system family) this may be the case when the reflection factor is about 65%. Due to the attenuation tolerances and thus small reflections of the components in the LC-TX-path, the measured reflection factor will not be 100% like it might be expected.

Check the 50 Ohm termination resistor of the Service Plug with a multi meter in case the reflected wave is out of specification. It has to be (50 +/- 2) Ohm. Larger differences, "open", or "shortcut" show that the resistor is defective

For the Service Plug kit, Avanto/ Espreet the other installed termination resistor can be used instead. Simply use a 90 degree connection adapter to connect the LC-TX-outlet to the other 50-Ohm-inlet (usually the Harmony/Symphony-"side"). Of course do not forget to check this resistor also

In case all contained termination resistors are defective (if more than one is built-in) send the Service Plug back for repair.

If the termination resistor does not show any malfunction, repeat the test or test the LC-TX-path with a transmit/receive coil (e.g. some Extremity Coils or the TxRx Head Coil).

This hint applies to all MR systems for which a Service Plug or in general a troubleshooting tool with a built-in 50 Ohm termination resistor is available.

2.9 Appendix - Service Plug

2.9.1 Description of Function Elements

- Frequency range 8 - 124 MHz
- 2x internal 50 Ohm power terminators for TX measurements, e.g. TestTools/RF/Cable for LC/TX:
 - TX1 for local coil plug #1 (Avanto)
 - TX2 for 4-channel coil plug (with available TX-signal)

Instead of the internal 50 Ohm resistors, external power terminators can be connected.

- 10V RX/Pre Amp voltage measurement:
 - without and with load (according to the preamplifier load)
 - latch enable/ disable (reset button for a new measurement).

LED-display according to number of channels:

- yellow LEDs: 10 V present but out of tolerance
- green LEDs: 10V within tolerance limits

Test sockets RX TP for DVM measurement of the 10V source

- Toggle switch for TX/RX mode with corresponding coil code CC/CD; default TX Mode. RX mode can be used for using the Service Plug as an RX coil (RF pick-up probe connected to TTX in). The Service Plug has two coil codes for TX and RX.
- Resistive attenuator array for checking RX cable from RCCS to PTAB and RCCS-tests using the TTX signal from modulator (with TestTools/RF/RCCS plug x)
- 108mA/-30V PIN diode (yellow/green) current/voltages measurement

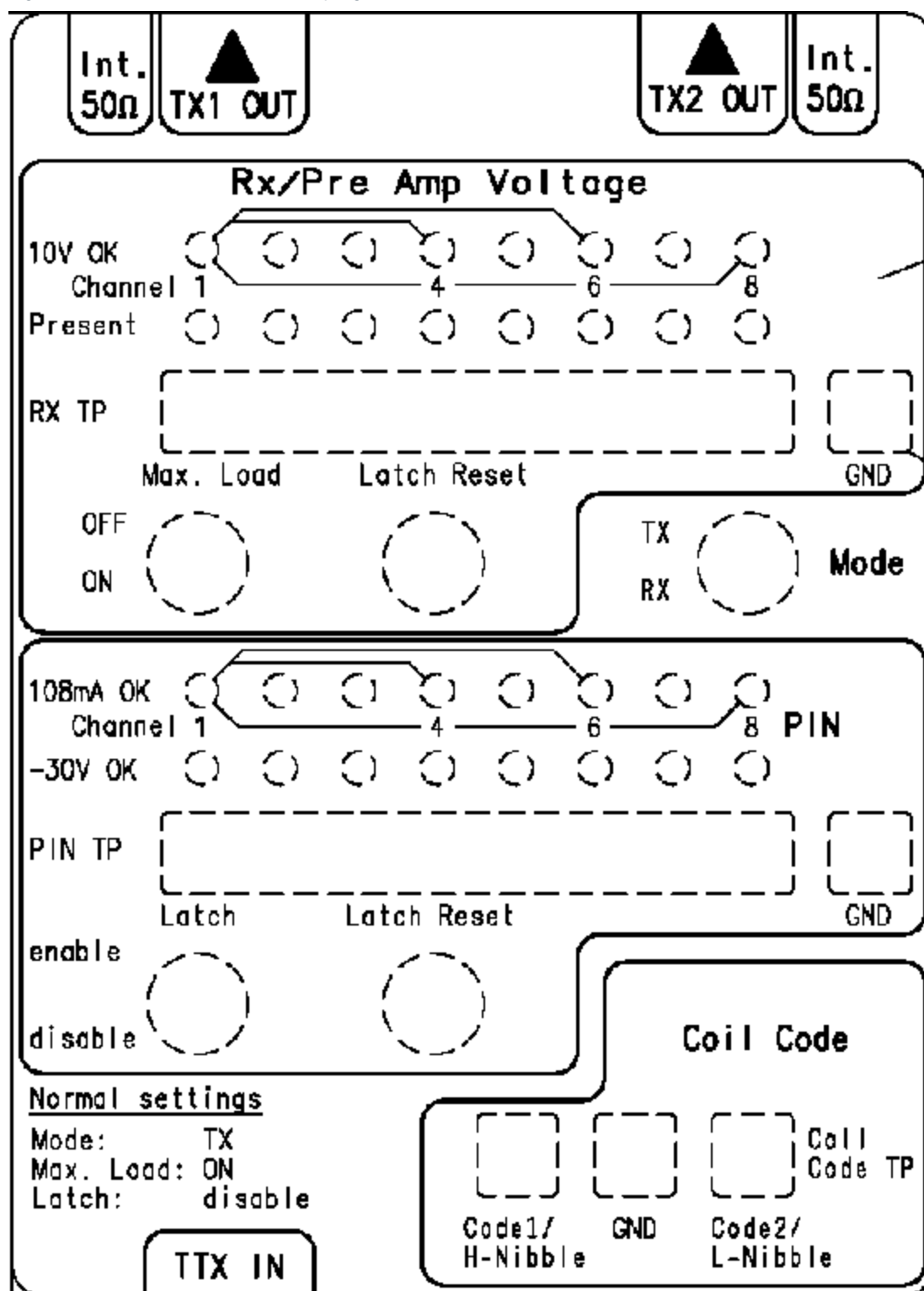
LED-display for ok status of PIN diodes control voltages

(used with TestTools/RFIS/Interactive)

Test sockets PIN TP for DVM measurements of I_F and U_R

- Checking the coil code detection (voltage). Approx. values for coil code "CC" is 7.8V for both sockets.

Fig. 52: Schematic view of service plug



2.9.1.1 Default Settings:

Mode: TX; Max.load: ON; Latch: disable

2.9.2 Notes on Use

2.9.2.1 General



Contrary to what is known from the other Tim systems, for MAGNETOM Avanto and Verio with dot Software it is necessary to switch the Service Plug to "RX".

If that is not done the following error message will be displayed at the magnet: #xx: wrong socket.



CAUTION

Never connect both the 6/8-channel plug and the 4-channel plug at the same time.

- » Irreversible damage to the Service Plug could result if the plugs are connected for a long time. A few seconds up to one minute is no problem.

- Never have external cables (e.g. for DVM, scope) connected when performing RF measurements with the Service Plug . The results would be incorrect.

2.9.2.2 RX/PreAMP-Voltage section

- After plug-in of the Service Plug, the latches have to be reset in case display of the LEDs is required. In case the latch of the PIN-section is disabled (normal setting), this is only necessary for the latch of the Rx/PreAmp-section.
- The 10V of RX channel #1 supplies the Rx/PreAmp section with power. When the channel has no voltage, the complete section will also be "dark".
- The measured values at the RX test points will change with the position of the Max.-Loaf switch by approx. 0.6V and depends on the length of the cables.

2.9.2.3 PIN section

- The 10V of RX channel 2 supplies the PIN-section with power. When RX channel 2 has no voltage, the complete PIN section will be "dark" as well.
- PIN_GND_1 is used as a reference ground.
- In bias current (I_F) mode, channels 1 and 2 will be dark if PIN_1 is missing. With PIN_GND_2, it will be channel 3 and 4; PIN_GND_3, channel 5 and 6; and with PIN_GND_4, channel 7 and 8.

2.9.2.4 Pick-up Coil

- Pick-up coils can be connected to TTX IN or RX-TP and GND (every channel, channel 1 is preferred).
- Pick-up coils connected to RX-TP and GND have to block the 10V coming from the RCCS either through an internal preamplifier or a capacitor. Otherwise, a connection between RX-TP and GND may result in a short-cut of the 10V.

- Pick-up coils connected to TTX IN do not need to have this, no 10V are available here. The disadvantage is that an additional attenuation (6-10 dB) has to be taken into account due to the Service Plug's internal structure.

Version 08

- Chapter (Procedures), Section (RFAS) - troubleshooting procedure for Tales 500 mV, extended to Avanto and Espree systems.(→ TALEs 500 mV Reference Voltage / Page 25)
- Minor editorial changes.

Version 09

- Chapter (Strategy) - section (→ Proposed sequence of test procedures - FRU-based / Page 6) added.
- All chapters - minor editorial changes.

Version 10

- Chapter (Procedures) - section (→ Check of 50 Ohm/Short Service Adapters (failed Tuning Calibration) / Page 74) added (CHARM TD_00015520).

There are no Hazard IDs in this document.

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